
Fitted Q Evaluation Without Bellman Completeness via Stationary Weighting

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Abstract

1 Fitted Q -evaluation (FQE) is a standard regression-based tool for off-policy evaluation,
2 but existing stability guarantees often rely on Bellman completeness, a strong
3 closure condition that can fail under function approximation. We study an alternative
4 route: changing the norm used in the regression step. The policy-evaluation
5 Bellman operator is contractive in the L^2 norm induced by the target policy's
6 stationary state-action distribution, whereas standard off-policy FQE projects Bellman
7 targets in the behavior-distribution norm. We propose stationary-weighted
8 FQE, which reweights each Bellman regression by the stationary target-to-behavior
9 density ratio. The method preserves FQE's modular supervised-learning form
10 while aligning the fitted projection with that contractive norm. We prove finite-
11 sample linear convergence to the stationary projected Bellman fixed point under
12 misspecification, without requiring Bellman completeness. The bound separates
13 finite-iteration, statistical, approximation, and weight-estimation errors, and shows
14 that ratio-estimation error is attenuated when the inherent Bellman error is small.
15 Controlled experiments show that stationary weighting can stabilize FQE and
16 reduce value error when behavior-norm regression overemphasizes regions rarely
17 visited by the target policy.

18 1 Introduction

19 Fitted Q -evaluation (FQE) is a standard regression-based method for policy evaluation in reinforcement
20 learning [9, 19, 21, 29, 41, 54, 55, 69]. Given offline transitions, each iteration forms one-step
21 Bellman targets for a fixed policy and regresses them onto a function class. This makes FQE
22 easy to implement, computationally stable, and compatible with off-the-shelf supervised learners
23 [2, 26, 42, 81]. However, under distribution shift, FQE can behave poorly with function approximation,
24 even under realizability and even in simple settings with linear function approximation [5, 29].
25 This instability arises because FQE iterates a projected Bellman operator [45, 52, 69], with the
26 projection norm determined by the sampling distribution [55]. The resulting projection can destroy
27 the contraction properties that make dynamic programming stable [6].

28 This practical challenge is reflected in both existing guarantees and hardness results. Positive guarantees
29 typically control the projection step through Bellman completeness or closely related closure
30 conditions, which require the Bellman image of each function in the class to remain in, or be well
31 approximated by, the class [20, 33, 55, 65]. This is a strong requirement for practical approximation
32 classes: restricted classes are rarely closed under Bellman updates, while enlarging the class to
33 improve closure can increase statistical complexity and does not by itself ensure completeness. Under
34 such conditions, the projection step is benign because the Bellman update is aligned with the fitted
35 class. Negative results show that this is not merely a proof artifact: without Bellman completeness
36 or comparable structural assumptions, off-policy evaluation with function approximation can be

statistically or computationally intractable in the worst case [4, 12, 25, 87]. Thus, the dependence on Bellman completeness is a central limitation of regression-based FQE.

This limitation has motivated a large literature on related fitted Bellman methods, including minimax and adversarial Q -learning methods based on saddle-point formulations of Bellman-residual minimization [13, 14, 23, 70, 73, 84, 86, 91]. These methods decouple the class used to estimate the value function from the class used to test Bellman residuals: rather than requiring the Bellman update to remain in the value class, they control residuals through a critic or discriminator class. This shifts the burden from Bellman completeness of the value class to assumptions on critic richness, coverage, and optimization [22, 34, 72, 91]. The resulting saddle-point problems are flexible, but introduce additional design and optimization choices, including the critic class, regularization, and primal-dual training procedure.

We take a complementary route: rather than replacing FQE’s regression-based update, we revisit the projection norm used in each Bellman regression. The evaluation Bellman operator is contractive in the L^2 norm induced by the target policy’s stationary distribution, whereas standard off-policy FQE projects Bellman targets in the behavior sampling norm. Under distribution shift, this mismatch can make the projected update unstable even when the underlying Bellman operator is contractive. This norm mismatch is a classical source of instability in projected policy evaluation [69] and motivates stabilizing corrections for temporal-difference learning [47, 60, 66]. For FQE, it suggests a simple intervention: reweight each Bellman regression by the target-to-behavior stationary density ratio, thereby projecting in the target stationary norm while preserving the supervised-learning form of the update.

We call the resulting procedure *stationary-weighted FQE*. It changes only the regression norm: the Bellman targets remain unchanged, no adversarial critic is introduced, and Bellman completeness is not imposed. This isolates a distinct source of FQE instability: not Bellman regression itself, but the norm in which projection errors are controlled. Under suitable coverage and weighting-error conditions, stationary-weighted FQE converges linearly to the stationary projected fixed point without requiring Bellman completeness.

Contributions.

1. **Stationary-weighted FQE.** We introduce stationary-weighted FQE, which reweights each Bellman regression by the target stationary-to-behavior state-action density ratio. This changes the projection norm while leaving the usual supervised regression update unchanged.
2. **Convergence without Bellman completeness.** We prove finite-sample convergence under function approximation without requiring Bellman completeness. The bound controls convergence to the stationary projected fixed point and separates finite-iteration, statistical, Bellman approximation, and density-ratio estimation errors.
3. **Robustness to approximate density ratios.** We characterize how imperfect density-ratio weights affect the fitted update. Their effect appears through a regression-curvature loss and an interaction with the Bellman approximation error, which vanishes as the class approaches Bellman completeness.

Scope and limitations. Our results isolate the role of the projection norm in FQE. This perspective is most useful when projected Bellman iteration is not already well controlled by linear, tabular, or completeness-based arguments [38, 50, 88]. The main practical limitations are coverage and density-ratio estimation: stationary weighting makes the overlap requirement explicit through the target stationary-to-behavior density ratio, whose stable estimation is the main added practical burden [36, 60, 72].

1.1 Related Work

Classical and minimax Bellman-error analyses. Classical analyses of fitted value iteration and fitted Q -evaluation typically control projected Bellman iteration in one of two ways. One route assumes that the projected Bellman operator is contractive in the regression norm, possibly with a degraded contraction factor [9, 29, 69]. A second route bounds how one-step approximation and estimation errors propagate across iterations [55, 65]. These guarantees require structure linking the Bellman operator, function class, and sampling distribution, such as Bellman completeness or

small inherent Bellman error, linear structure, learned representations, or state-space discretization [11, 18, 38, 50, 62, 80, 85, 88].

A related line replaces least-squares Bellman regression with minimax, adversarial, or pessimistic Bellman-error objectives [3, 16, 37, 70, 71, 84]. These methods test Bellman residuals against auxiliary critic or test-function classes and obtain guarantees under conditions such as partial coverage, dual realizability, or critic richness. This contrasts with classical FQE analyses based on Bellman completeness, which handle distribution shift through concentrability or distribution-mismatch constants [20, 55, 65, 87]. Our approach is complementary: we keep the least-squares Bellman regression form of FQE and instead change the projection norm. In some minimax formulations, the auxiliary class represents occupancy-ratio or marginalized importance-weight functions [70, 73, 84]. Stationary-weighted FQE uses stationary ratios directly to define the regression norm. Rather than relying on an auxiliary critic class to detect the relevant Bellman residuals, our analysis requires the stationary density ratio to be estimated accurately enough for the fitted projection to approximate the target-stationary projection.

Weighted temporal-difference and density-ratio methods. Emphatic TD and related weighted TD methods stabilize off-policy bootstrapping by choosing update weights designed to improve the stability or contraction properties of projected Bellman updates [31, 47, 60, 63, 66, 89, 90]. These results are closest to ours in spirit, but they study TD-style stochastic approximation with tailored update weights rather than regression-based fitted Bellman updates. In the linear, discrete-state setting, Van Roy [80] similarly showed that using invariant distributions of relevant policies as projection weights can sharpen approximate value-iteration bounds and related these projection norms to TD learning.

Our setting differs from weighted TD in three ways. First, the weights define the least-squares projection norm in each FQE regression, rather than update weights in a stochastic-approximation recursion. Second, our guarantees are formulated for fitted regression with possibly nonlinear function classes and general state spaces, whereas much of the classical emphatic-TD theory is linear or finite-state. Third, we separate the ideal target-stationary projection from its estimated implementation: our guarantees assume contraction under ideal stationary weighting and quantify the additional error from approximate density ratios. This error is smaller when the function class is closer to Bellman complete.

The stationary density-ratio weights used by our method can be estimated with stationary or discounted ratio-estimation methods, including DICE-style and minimax estimators for occupancy corrections in off-policy evaluation and learning [30, 46, 56, 70, 72, 92]. Stable weight estimation is especially delicate with nonlinear approximation or limited overlap [49, 60, 72, 93]. We use these estimators as an implementation device, while our analysis focuses on how their approximation error affects fitted Bellman regression under misspecification.

1.2 Preliminaries

We consider a discounted MDP with state space $\mathcal{S}$, finite action space $\mathcal{A}$, transition kernel $P(\cdot \mid s, a)$, reward $r_0(s, a)$, and discount $\gamma \in [0, 1)$. A fixed target policy $\pi(a \mid s)$ induces the state-action transition operator

$$(P_\pi Q)(s, a) := E[Q(S', A') \mid S = s, A = a], \quad S' \sim P(\cdot \mid s, a), \quad A' \sim \pi(\cdot \mid S').$$

The policy-evaluation Bellman operator and its fixed point are

$$\mathcal{T}Q := r_0 + \gamma P_\pi Q, \quad Q^\pi = \mathcal{T}Q^\pi.$$

The fixed point Q^π is the target state-action value function: the expected discounted return after taking action a in state s and following π thereafter.

We observe i.i.d. transitions $\mathcal{D}_n = \{(S_i, A_i, R_i, S'_i)\}_{i=1}^n$, where $(S, A) \sim \nu_b$, $S' \sim P(\cdot \mid S, A)$, and $R = r_0(S, A) + \varepsilon$ with $E[\varepsilon \mid S, A] = 0$. Here ν_b is the behavior distribution on $\mathcal{S} \times \mathcal{A}$. We use i.i.d. transition sampling for simplicity; trajectory data can be handled under mixing conditions.

For any state-action distribution ν , write

$$(\pi f)(s) := \sum_{a \in \mathcal{A}} \pi(a \mid s) f(s, a), \quad \|f\|_{2, \nu} := \left(\int f(s, a)^2 \nu(ds, da) \right)^{1/2}.$$

Let μ_π denote a stationary state-action distribution for the Markov chain induced by (π, P) , i.e., $\mu_\pi f = \mu_\pi P_\pi f$. By convention, $0/0 = 0$.

1.3 Standard FQE and Norm Mismatch

When the model is known, classical dynamic programming computes the fixed point Q^π by iterating the Bellman operator $\mathcal{T}$ [6, 7]. Since $\mathcal{T}$ is a γ -contraction in sup norm, these iterates converge geometrically to Q^π . With function approximation and offline data, FQE replaces exact Bellman iteration by projected Bellman iteration [19, 41, 55]. Given a class $\mathcal{F}$, its empirical update is

$$\widehat{Q}^{(k+1)} \in \arg \min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n (R_i + \gamma (\pi \widehat{Q}^{(k)})(S'_i) - f(S_i, A_i))^2.$$

Thus each step forms one-step Bellman targets and projects them onto $\mathcal{F}$ by least squares. In the population limit, FQE iterates

$$\mathcal{T}_{\mathcal{F}}^{\nu_b} := \Pi_{\mathcal{F}}^{\nu_b} \mathcal{T},$$

where $\Pi_{\mathcal{F}}^{\nu_b}$ is the $L^2(\nu_b)$ -projection onto $\mathcal{F}$, and ν_b is the behavior state-action distribution.

If $\mathcal{F}$ is Bellman complete, meaning $\mathcal{T}f \in \mathcal{F}$ for all $f \in \mathcal{F}$, the projection is benign: $\Pi_{\mathcal{F}}^{\nu_b} \mathcal{T}f = \mathcal{T}f$ on $\mathcal{F}$. The fitted update therefore coincides with the Bellman update and inherits its sup-norm contraction. Classical FQE analyses use Bellman completeness, or related one-step approximation conditions, to control how approximation and statistical errors propagate across iterations [12, 55].

Without Bellman completeness, this argument breaks: FQE iterates $\mathcal{T}_{\mathcal{F}}^{\nu_b}$, and the projection norm can affect stability. The projection $\Pi_{\mathcal{F}}^{\nu_b}$ is nonexpansive in $L^2(\nu_b)$, whereas the policy-evaluation Bellman operator contracts in a different L^2 norm, namely $L^2(\mu_\pi)$, where μ_π is the stationary state-action distribution of the target policy [69]. Under distribution shift, these norms can be misaligned, so the behavior-projected update need not be contractive. Consequently, FQE may converge slowly, converge to a behavior-projected solution far from Q^π , or fail to converge [15, 60, 69].

1.4 Resolving Norm Mismatch

We resolve the mismatch by projecting in the stationary-distribution norm in which the target-policy Bellman operator is contractive. Under on-policy stationary sampling, FQE already uses this norm: its least-squares projection is taken in $L^2(\mu_\pi)$, where the Bellman operator contracts [31, 60, 69]. Off policy, we recover the same population projection by reweighting each Bellman regression so that its loss corresponds to sampling from μ_π .

For a class $\mathcal{F} \subseteq L^2(\mu_\pi)$, define the stationary projected Bellman operator

$$\mathcal{T}_{\mathcal{F}} := \Pi_{\mathcal{F}} \mathcal{T}, \quad \Pi_{\mathcal{F}} g := \arg \min_{f \in \mathcal{F}} \|f - g\|_{2, \mu_\pi}, \quad (1)$$

which is well defined under the following conditions.

C1 Stationarity. The Markov chain induced by (π, P) has a stationary state-action distribution μ_π .

C2 Closed and convex class. The class $\mathcal{F}$ is closed and convex in $L^2(\mu_\pi)$.

Lemma 1 (Stationary projected Bellman contraction [69]). *Assume C1–C2. Then, for all $Q, Q' \in \mathcal{F}$,*

$$\|\mathcal{T}_{\mathcal{F}} Q - \mathcal{T}_{\mathcal{F}} Q'\|_{2, \mu_\pi} \leq \gamma \|Q - Q'\|_{2, \mu_\pi}.$$

Condition C1 holds under standard ergodicity assumptions [64]. The closed convexity part of Condition C2 makes $\Pi_{\mathcal{F}}$ well defined and nonexpansive in $L^2(\mu_\pi)$.

2 Stationary-Weighted Fitted Q-Evaluation

2.1 Algorithm

We modify only the regression step in FQE. Each Bellman target is fit by weighted least squares, with weights chosen so that the population loss is the $L^2(\mu_\pi)$ loss rather than the behavior-distribution

171 loss. We present the algorithm using the stationary state-action density ratio; Appendix B.2 extends
172 the construction to other admissible weights.

173 Assume $\mu_\pi \ll \nu_b$ and define the stationary state-action density ratio $w_\pi(s, a) := \frac{d\mu_\pi}{d\nu_b}(s, a)$. Given
174 an estimate $\hat{w}_\pi$, stationary-weighted FQE updates

$$\hat{Q}^{(k+1)} \in \arg \min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \hat{w}_\pi(S_i, A_i) \left\{ R_i + \gamma(\pi \hat{Q}^{(k)})(S'_i) - f(S_i, A_i) \right\}^2.$$

175 If $\hat{w}_\pi = w_\pi$, the population update is exactly the $L^2(\mu_\pi)$ -projected Bellman update.

176 Relative to standard FQE, the only structural change is the regression weight. The method is therefore
177 compatible with generic supervised learners, including neural networks and boosted ensembles. Its
178 main additional cost is estimating the stationary state-action density ratio, which can be done once
179 before the fitted iteration.

180 2.2 Estimation of Stationary Ratios

181 Stationary-weighted FQE requires weights approximating $w_\pi = d\mu_\pi/d\nu_b$. Estimation of stationary
182 or discounted occupancy ratios is well studied in off-policy evaluation, including classical stationary-
183 distribution estimators [27, 48, 61], DICE-style saddle-point methods [43, 44, 46, 56, 57, 92],
184 minimax weight estimation [70, 72], and projected or balancing-weight methods [3, 83].

185 Stationary ratios are the undiscounted analogues of discounted occupancy ratios, which appear in
186 importance-sampling, doubly robust, and efficient policy-value estimators [35, 39, 40, 67, 76]. Under
187 suitable ergodicity conditions, stationary ratios arise as the $\gamma \rightarrow 1$ limit of discounted occupancy
188 ratios. Thus, stationary-weighted FQE uses an auxiliary ratio estimate already common in off-policy
189 evaluation, but uses it to stabilize Bellman regression rather than only to debias the final value
190 estimate.

191 The key identifying condition is stationarity. Since $w_\pi \nu_b = \mu_\pi$, the ratio satisfies

$$E[w_\pi(S, A) \{g(S, A) - g(S', A')\}] = 0 \quad \text{for all } g,$$

192 where $(S, A) \sim \nu_b$, $S' \sim P(\cdot \mid S, A)$, and $A' \sim \pi(\cdot \mid S')$. Minimax estimators enforce this
193 condition over a critic class while searching over a ratio class, with a normalization constraint such as
194 $E_{\nu_b} w_\pi = 1$; see Appendix C. For linear or RKHS critics, the inner maximization can often be solved
195 in closed form [17, 59, 70, 83], yielding computationally simple weights. Section 3 quantifies how
196 ratio-estimation error affects stationary-weighted FQE.

197 3 Theoretical Guarantees

198 We analyze the ERM version of stationary-weighted FQE in Algorithm 1. The proof follows the
199 inexact Picard-iteration framework of Munos and Szepesvári [55], but adds a one-step regression
200 bound for estimated weights. The key issue is that contraction holds for the ideal stationary projection,
201 whereas the algorithm uses empirical regressions with approximate weights.

202 For clarity, we focus on stationary density-ratio weighting. The same argument applies to any admis-
203 sible weight w_π that preserves the stationary projected Bellman update; under Bellman completeness,
204 any weight is admissible because the Bellman residual is zero (Appendix B.2).

205 More generally, the proof applies to any limiting weighted projection that is contractive, with the
206 corresponding contraction factor. This includes discounted occupancy ratios used as lower-variance
207 approximations to stationary ratios (Appendix B.4), at the cost of a weaker contraction factor. Unlike
208 stationary ratios, discounted occupancy ratios can be defined without ergodicity. Moment-matched
209 weights over a restricted critic class may also improve contraction without recovering the exact
210 stationary ratio, by emphasizing state-action regions relevant to the target policy [60].

211 3.1 Projected Fixed Point and Inexact Iteration

212 The population target of stationary-weighted FQE is the fixed point $Q_{\mathcal{F}}^\pi$ of the stationary projected
213 Bellman operator

$$\mathcal{T}_{\mathcal{F}} Q := \Pi_{\mathcal{F}} \mathcal{T} Q = \arg \min_{f \in \mathcal{F}} \|\mathcal{T} Q - f\|_{2, \mu_\pi}.$$

By Lemma 1, $\mathcal{T}_{\mathcal{F}}$ is a γ -contraction in $L^2(\mu_{\pi})$, so $Q_{\mathcal{F}}^{\pi}$ exists and is unique. If $Q^{\pi} \in \mathcal{F}$, then $Q_{\mathcal{F}}^{\pi} = Q^{\pi}$; otherwise, the next result bounds its distance from Q^{π} by the best stationary-norm approximation error.

Theorem 1 (Function approximation error). *Under Conditions C1–C2,*

$$\|Q_{\mathcal{F}}^{\pi} - Q^{\pi}\|_{2, \mu_{\pi}} \leq \frac{1}{1 - \gamma} \inf_{f \in \mathcal{F}} \|f - Q^{\pi}\|_{2, \mu_{\pi}}.$$

The algorithmic error is controlled by the standard inexact Picard recursion.

Lemma 2 (Inexact Picard iteration). *Assume Conditions C1–C2. If $\|\widehat{Q}^{(k)} - \mathcal{T}_{\mathcal{F}}\widehat{Q}^{(k-1)}\|_{2, \mu_{\pi}} \leq \eta_k$ for all $k \geq 1$, then, for every $K \geq 1$,*

$$\|\widehat{Q}^{(K)} - Q_{\mathcal{F}}^{\pi}\|_{2, \mu_{\pi}} \leq \gamma^K \|\widehat{Q}^{(0)} - Q_{\mathcal{F}}^{\pi}\|_{2, \mu_{\pi}} + \sum_{j=1}^K \gamma^{K-j} \eta_j.$$

3.2 One-step Regression Error under Approximate Weighting

To prove the main result, we combine Lemma 2 with a high-probability bound on $\|\widehat{Q}^{(k)} - \mathcal{T}_{\mathcal{F}}\widehat{Q}^{(k-1)}\|_{2, \mu_{\pi}}$, the error from one empirical weighted Bellman regression. This is where statistical error and approximate weighting enter. We first state the required conditions, then give the main technical lemma.

We state the ERM rates in terms of metric-entropy critical radii. For $\varepsilon > 0$, let $N(\varepsilon, \mathcal{F}, L^2(\mathbb{Q}))$ denote the ε -covering number of $\mathcal{F}$ under $L^2(\mathbb{Q})$, and define

$$\mathcal{J}(\delta, \mathcal{F}) := \int_0^{\delta} \sup_{\mathbb{Q}} \sqrt{\log N(\varepsilon, \mathcal{F}, L^2(\mathbb{Q}))} d\varepsilon, \quad \delta_n := \inf \left\{ \delta > 0 : \frac{\mathcal{J}(\delta, \mathcal{F})}{\sqrt{n} \delta^2} \leq 1 \right\},$$

where the supremum is over finitely supported distributions $\mathbb{Q}$. The critical radius δ_n gives the usual localized ERM rates for standard classes, including $\sqrt{s \log(p/s)/n}$ for sparse linear classes, $\sqrt{V \log(n/V)/n}$ for VC-subgraph classes, and $n^{-\alpha/(2\alpha+d)}$ for d -variate Hölder or Sobolev classes of smoothness α [51, 58, 79, 82]. We impose the following weighted ERM conditions.

C3 Coverage and bounded weights. $\mu_{\pi} \ll \nu_b$ and there exist $1 \leq \kappa_{\text{cov}} < \infty$ such that

$$\|w_{\pi}\|_{\infty} \leq \kappa_{\text{cov}}, \quad \|\widehat{w}_{\pi}\|_{\infty} \leq \kappa_{\text{cov}}, \quad \|(\widehat{w}_{\pi} - w_{\pi})/\sqrt{w_{\pi}}\|_{\infty} \leq \kappa_{\text{cov}}.$$

C4 Bounded regression objects. There exists $M < \infty$ such that $|R| \leq M$ and $\sup_{f \in \mathcal{F}} \|f\|_{\infty} \leq M$.

C5 Sample splitting. The weight estimator $\widehat{w}_{\pi}$ is computed from data independent of $\mathcal{D}_n$.

C6 Entropy regularity. The class satisfies $\mathcal{J}(\infty, \mathcal{F}) < \infty$ and $\frac{\mathcal{J}(\delta, \mathcal{F})}{\delta \sqrt{\log \log(1/\delta)}} \rightarrow \infty$ as $\delta \downarrow 0$.

Condition C3 imposes bounded target-behavior coverage and keeps the estimated weights controlled. It is the analogue of coverage or concentrability assumptions in offline RL [87], and is related to dual realizability conditions in minimax and saddle-point approaches to offline policy evaluation [70]. The boundedness requirements in Conditions C3 and C4 could be relaxed under appropriate tail assumptions, such as sub-exponential norms. Condition C5 separates the weight-estimation and Bellman-regression samples; it can be replaced by cross-fitting or by complexity control for $\widehat{w}_{\pi}$ [24, 75]. Finally, Condition C6 controls localized Rademacher complexities, following Munos and Szepesvári [55], and holds for classes with polynomial metric entropy.

Define the fitted difference class and the relative distortion of the weighted quadratic norm by

$$\mathcal{H}_{\mathcal{F}} := \{f - f' : f, f' \in \mathcal{F}\}, \quad \rho_{\mathcal{H}} := \sup_{h \in \mathcal{H}_{\mathcal{F}}} \left| \frac{E_{\nu_b}[(\widehat{w}_{\pi} - w_{\pi})h^2]}{E_{\nu_b}[w_{\pi}h^2]} \right|.$$

C7 Weighted-loss curvature stability. Assume $\rho_{\mathcal{H}} < 1$.

246 This condition requires the estimated-weight quadratic form to remain comparable to the ideal
247 weighted quadratic form on fitted directions: for every $h \in \mathcal{H}_{\mathcal{F}}$,

$$E_{\nu_b}[\hat{w}_\pi h^2] \geq (1 - \rho_{\mathcal{H}})E_{\nu_b}[w_\pi h^2].$$

248 It is implied by the pointwise relative-error bound $\|\hat{w}_\pi/w_\pi - 1\|_\infty < 1$. It also follows from
249 $B_{\mathcal{H}}\|\hat{w}_\pi/w_\pi - 1\|_{2,w_\pi\nu_b} < 1$, where $B_{\mathcal{H}} := \sup_{h \in \mathcal{H}_{\mathcal{F}}} \|h\|_{4,w_\pi\nu_b}^2 / \|h\|_{2,w_\pi\nu_b}^2$ measures the spikiness
250 of fitted directions. Hence, if $\|\hat{w}_\pi/w_\pi - 1\|_\infty \wedge B_{\mathcal{H}}\|\hat{w}_\pi/w_\pi - 1\|_{2,w_\pi\nu_b} = o_p(1)$, then $\rho_{\mathcal{H}} = o_p(1)$,
251 so Condition C7 holds with probability tending to one.

252 The one-step analysis uses two error terms. Define the weight-error-weighted inherent Bellman error
253 [55] and the statistical regression error by

$$\omega_{\text{Bell},w} := \sup_{Q \in \mathcal{F}} \left\| \frac{\hat{w}_\pi - w_\pi}{\sqrt{w_\pi}} (\mathcal{T}Q - \mathcal{T}_{\mathcal{F}}Q) \right\|_{2,\nu_b}, \quad \varepsilon_{\text{stat}}(m, \tau) := \kappa_{\text{cov}} \left(\delta_n + \sqrt{\frac{\log(m/\tau)}{n}} \right).$$

254 For unweighted FQE, $\hat{w}_\pi \equiv 1$, so this term measures the interaction between the fixed projection-
255 norm mismatch $1 - w_\pi$ and the inherent Bellman residual. Stationary weighting replaces this fixed
256 mismatch by the weight-estimation error $\hat{w}_\pi - w_\pi$: approximate weights are therefore less harmful
257 when $\mathcal{T}Q - \mathcal{T}_{\mathcal{F}}Q$ is small in regions where the weight error is large.

258 **Lemma 3** (One-step weighted regression error). *Assume Conditions C1–C7. There exists $C =$
259 $C(M) < \infty$ such that, for any $\tau \in (0, 1)$, with probability at least $1 - \tau$,*

$$\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{(k)}) - \hat{Q}^{(k+1)}\|_{2,\mu_\pi} \leq \frac{C}{1 - \rho_{\mathcal{H}}} \{\varepsilon_{\text{stat}}(1, \tau) + \omega_{\text{Bell},w}\}.$$

260 The first term is the usual ERM error. The second term captures weight error interacting with the
261 Bellman approximation residual $\mathcal{T}Q - \mathcal{T}_{\mathcal{F}}Q$; it is therefore small when the weights are accurate or
262 when the inherent Bellman error is small. The factor $(1 - \rho_{\mathcal{H}})^{-1}$ is the curvature loss from using
263 estimated rather than oracle weights in the fitted regression.

264 3.3 Main Finite-Sample Guarantee

265 The main result follows by combining the one-step regression bound (Lemma 3) with the deterministic
266 recursion (Lemma 2).

267 **Theorem 2** (Finite-sample error bound for stationary-weighted FQE). *Assume Conditions C1–C7.*
268 *There exists $C = C(M) < \infty$ such that, for all $\tau \in (0, 1)$ and $K \in \mathbb{N}$, with probability at least*
269 *$1 - \tau$,*

$$\|\hat{Q}^{(K)} - Q_{\mathcal{F}}^\pi\|_{2,\mu_\pi} \leq \gamma^K \|\hat{Q}^{(0)} - Q_{\mathcal{F}}^\pi\|_{2,\mu_\pi} + \frac{C}{(1 - \gamma)(1 - \rho_{\mathcal{H}})} \{\varepsilon_{\text{stat}}(K, \tau) + \omega_{\text{Bell},w}\}.$$

270 Theorem 2 is an FQE analogue of classical inexact Bellman-iteration bounds [55], stated in the
271 stationary $L^2(\mu_\pi)$ norm and without Bellman completeness. The first term is the finite-iteration bias.
272 The remaining terms are the one-step regression error and the interaction between weight error and
273 Bellman incompleteness. These terms are inflated by the curvature-loss factor $(1 - \rho_{\mathcal{H}})^{-1}$ when
274 oracle weights are replaced by estimated weights; with oracle weights, $\rho_{\mathcal{H}} = 0$ and the weight-error
275 interaction vanishes.

276 **Errors are controlled in the stationary norm.** Theorem 2 controls FQE error in $L^2(\mu_\pi)$, the norm
277 induced by the target policy’s long-run state-action distribution, rather than under the data-collection
278 distribution. Thus, errors are measured where the target policy spends time. This directly controls
279 downstream policy-evaluation error: the policy value under the stationary state distribution ρ_π is
280 $J(Q) := E_{\rho_\pi}\{(\pi Q)(S)\} = E_{\mu_\pi}\{Q(S, A)\}$, and therefore $|J(\hat{Q}) - J(Q)| \leq \|\hat{Q} - Q\|_{2,\mu_\pi}$. No
281 additional behavior-to-target norm conversion is needed.

282 **Approximate completeness attenuates weight error.** Apart from the curvature factor $(1 - \rho_{\mathcal{H}})^{-1}$,
283 the only term through which weight-estimation error enters the bound is $\omega_{\text{Bell},w}$. This is a weight-
284 error-weighted analogue of the inherent Bellman error of Munos and Szepesvári [55]. To see this,
285 let $\varepsilon_{\text{Bell}} := \sup_{Q \in \mathcal{F}} \|\mathcal{T}Q - \mathcal{T}_{\mathcal{F}}Q\|_{2,\mu_\pi}$ and $\varepsilon_w^2 := E_{\nu_b}[(\hat{w}_\pi - w_\pi)^2/w_\pi]$. If $\|\mathcal{T}Q - \mathcal{T}_{\mathcal{F}}Q\|_\infty \leq$
286 $L_{\mathcal{R}}\|\mathcal{T}Q - \mathcal{T}_{\mathcal{F}}Q\|_{2,\mu_\pi}^\beta$ for all $Q \in \mathcal{F}$, with $\beta \in [0, 1]$, then $\omega_{\text{Bell},w} \leq L_{\mathcal{R}}\varepsilon_w\varepsilon_{\text{Bell}}^\beta$. Thus approximate

287 Bellman completeness reduces the effect of imperfect weights through the residual-interaction term,
 288 while the fitted-regression geometry is controlled separately by $\rho_{\mathcal{H}}$.

289 **Instantiating weight accuracy.** The weight error ε_w can be controlled by standard ratio-estimation
 290 guarantees [70, 72]. Let $\mathcal{W} = \{w_\theta : \theta \in \Theta\}$ be a ratio class and $\mathcal{G}$ a critic class. On an independent
 291 ratio-estimation sample of size n_w , let P_{n_w} denote the empirical average over (S, A, S', A') , with
 292 $(S, A) \sim \nu_b$, $S' \sim P(\cdot | S, A)$, and $A' \sim \pi(\cdot | S')$. A representative estimator is

$$\hat{w}_\pi \in \arg \min_{w \in \mathcal{W}} \sup_{g \in \mathcal{G}} \left\{ P_{n_w} [w(S, A) \{g(S, A) - g(S', A')\}] - \frac{1}{2} P_{n_w} [g(S, A)^2] \right\} \quad \text{s.t. } P_{n_w} w = 1.$$

293 For finite-dimensional linear classes, $w_\theta(s, a) = \theta^\top \psi(s, a)$ and $g_\beta(s, a) = \beta^\top \phi(s, a)$, Appendix H
 294 shows that, under bounded features, overlap $\underline{d} \leq w_\pi$, and restricted identification of stationarity
 295 moments with constant ζ , up to critic-class defect $\eta_{\mathcal{G}}$, with probability at least $1 - \tau$,

$$\varepsilon_w \lesssim \underline{d}^{-1/2} \left\{ \inf_{\theta \in \Theta} \|w_\theta - w_\pi\|_{2, \nu_b} + \eta_{\mathcal{G}} + \zeta^{-1} \sqrt{\frac{p + m + \log(1/\tau)}{n_w}} \right\}.$$

296 Here $\eta_{\mathcal{G}}$ is the approximation defect from restricting to the critic class $\mathcal{G}$, and ζ is a restricted singular-
 297 value constant for the stationarity moment matrix. Thus the bound has the usual finite-dimensional
 298 minimax rate when the ratio class is well specified, the critic class captures the relevant stationarity
 299 violations, and the moment matrix is well conditioned [8, 17, 70, 72].

300 4 Experimental investigation

301 We use three controlled experiments to test the mechanism suggested by the theory. Two finite-state
 302 diagnostics isolate norm mismatch: Baird’s seven-state example [5] and sparse Garnet MDPs with
 303 realizability but not Bellman completeness. With known stationary ratios, stationary-weighted FQE
 304 remains stable as overlap weakens, while unweighted FQE slows or diverges; see Figure 1A–B
 305 and Appendix D.1. The main continuous linear-Gaussian benchmark then tests whether occupancy
 306 weighting improves the target-stationary accuracy of fitted Q -functions with estimated weights.

307 **Controlled continuous benchmark.** Our main continuous experiment uses a two-dimensional
 308 linear-Gaussian controlled Markov process,

$$S_{t+1} = BS_t + CA_t + \epsilon_t, \quad r(s, a) = -\|s - s^*\|^2 - \lambda a^2,$$

309 with Gaussian linear behavior and target policies. The target gain is fixed, and the behavior gain
 310 is shifted away from it to vary off-policy mismatch. For each shift and seed, we draw $n = 4000$
 311 independent one-step transitions from the stationary behavior state-action distribution. We compute
 312 Q^π , V^π , and the policy value analytically, and evaluate Q -error under the target stationary distribution.
 313 FQE uses discount $\gamma = 0.95$. Estimated stationary weights are fit from the same offline transitions
 314 used for FQE.

315 The experiment uses a deliberately misspecified affine Q -class. We compare standard FQE with
 316 stationary-weighted FQE using oracle weights, unregularized RBF moment weights, exponential-
 317 quadratic moment weights, and CV-Tikhonov RBF moment weights. The RBF ratios are estimated
 318 in a standardized finite-dimensional RBF feature class by solving the stationarity minimax moment
 319 equations described in Section 3.3 [70]; in this benchmark, the resulting estimator is closed form.
 320 The exponential-quadratic model sets $w_\theta(s, a) = \exp\{\theta^\top \psi(s, a)\}$, where ψ contains quadratic
 321 state-action features. This log-ratio class is correctly specified in the linear-Gaussian setting. The
 322 unregularized RBF estimator uses no Tikhonov regularization or truncation, whereas the CV-Tikhonov
 323 estimator selects the ridge parameter by five-fold held-out stationarity moment risk, as motivated by
 324 finite-sample theory [72].

325 Figure 1C plots target-stationary Q -error against the oracle stationary-ratio effective sample size
 326 (ESS) fraction, $\text{ESS}(w)/n = (\sum_i w_i)^2 / (n \sum_i w_i^2)$, with behavior shift shown on the top axis.
 327 Weighting has little effect near on-policy, where the behavior and target stationary distributions are
 328 close. As overlap worsens, stationary weighting reduces target-stationary Q -error, with the largest
 329 gains obtained by oracle weights and by the correctly specified exponential-quadratic ratio model.
 330 Figure 1D reports ESS and tail-weight diagnostics for the oracle and estimated weights; additional
 331 stabilizers are reported in Appendix D.

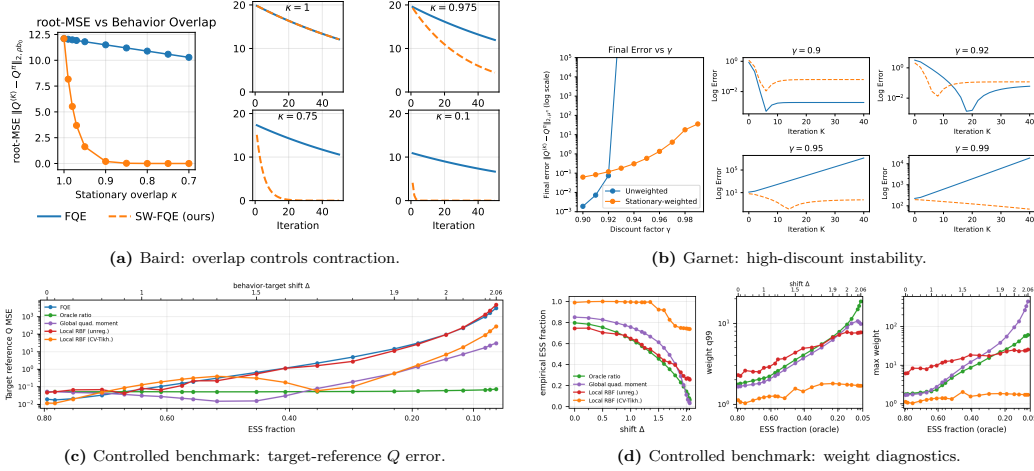


Figure 1: **A–B:** finite-state norm-mismatch examples. Stationary weighting restores fast decay in Baird’s example and prevents the high-discount divergence seen in Garnet MDPs. **C–D:** continuous linear-Gaussian simulation. Panel C plots target-stationary Q -error against the oracle stationary-ratio effective sample size (ESS) fraction, with behavior–target shift shown on the top axis. Stationary weighting reduces target-stationary Q -error as overlap deteriorates. Panel D reports empirical ESS and tail-weight diagnostics for oracle and estimated weights.

The results illustrate the bias–variance tradeoff predicted by the theory. Unweighted FQE is stable, but under behavior–target shift it fits the misspecified affine Q -class in behavior-relevant regions, leading to larger target-stationary error. Oracle weighting removes this projection mismatch and gives the best attainable correction within the same Q -class. The unregularized RBF moment estimator is unstable under limited coverage, producing nonuniform weights that poorly approximate the oracle ratio. CV-Tikhonov regularization gives a lower-variance correction and improves on unweighted FQE under moderate-to-severe shift, but remains biased when overlap is poor. The exponential-quadratic model performs best among estimated weights because its log-ratio class is correctly specified in this benchmark, although its lower ESS and larger maximum weights show the variance cost of ratio extrapolation.

We also compare against a closed-form minimax Bellman-residual estimator using the same affine Q -class and an RBF critic. For both estimated weighting and minimax fitting, Figure 3 in Appendix D.3 reports deployable CV-tuned rules and oracle-tuned diagnostics. The CV rules choose Tikhonov regularization by held-out stationarity or Bellman moment risk, without using oracle target-stationary error. Minimax fitting is competitive under favorable tuning, but does not uniformly dominate stationary weighting in the deployable finite-sample setting. This comparison suggests that stationary weighting is best viewed as complementary to minimax Bellman-residual fitting: it preserves the regression structure of FQE while targeting the norm in which policy evaluation contracts.

5 Conclusion

We studied FQE without Bellman completeness through a norm-mismatch lens: standard off-policy FQE projects Bellman targets in the behavior norm, whereas policy evaluation contracts in the target stationary norm. Stationary-weighted FQE changes only the regression weights, preserving FQE’s supervised-learning structure while aligning the fitted projection with this contraction. Our finite-sample bound gives linear convergence to the stationary projected fixed point under misspecification and off-policy sampling, separating finite-iteration, statistical, Bellman approximation, and density-ratio estimation errors. The result shows that Bellman completeness is not the only way to control fitted iteration: one can instead align the projection norm with the distribution under which policy evaluation is stable. The main limitations are coverage and stable density-ratio estimation; Appendix A discusses these limitations, connections to minimax Bellman methods, and future directions.

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A Expanded Conclusion

We studied FQE without Bellman completeness through a norm-mismatch lens: standard off-policy FQE projects Bellman targets in the behavior norm, whereas the target-policy Bellman operator contracts in the stationary norm. Stationary-weighted FQE changes only the regression weights, preserving the supervised-learning structure of FQE while aligning the projection norm with this contraction. Our finite-sample bound gives linear convergence to the stationary projected fixed point under misspecification and off-policy sampling, separating finite-iteration, statistical, Bellman approximation, and density-ratio estimation errors. The analysis further shows that approximate density ratios affect FQE through curvature loss and a residual-interaction term scaled by the inherent Bellman error.

The main practical limitations are coverage and density-ratio estimation. Poor overlap can make exact stationary ratios unstable, suggesting a role for discounted or regularized occupancy weights and stable balancing estimators in limited-coverage regimes. Stationary weighting should therefore be viewed as complementary to fitted Bellman-error minimization and minimax Q -learning methods, rather than as a replacement. Minimax objectives provide a powerful way to control Bellman residuals and can perform well with suitable tuning, while stationary weighting offers a simple regression-based route for aligning the fitted objective with the target-policy distribution. Because stationary ratios enter through regression weights, approximate ratios can still be useful when they move the fitted norm toward the target stationary distribution. In our continuous benchmark, regularized estimated weights improve performance in lower-overlap regimes and, when the correction is imperfect, remain close to unweighted FQE.

Overall, stationary reweighting is a minimal modification of FQE that restores the contraction geometry while making the usual coverage requirements explicit. Understanding how stationary weighting and minimax objectives can be combined, and how to tune them reliably under limited overlap, is an important direction for future work.

Organization. The appendix is organized as follows. Appendix B.3 discusses the regression-geometry condition and the residual interpolation bound, and Appendix B.4 records consequences for approximate weights. Appendix B.5 treats implicit stationary sampling with estimated rewards. Appendix C describes stationary-ratio estimation, Appendix D gives experimental details, Appendix E collects contraction results, and the final two sections prove the technical and main results.

B Additional Details

B.1 Algorithm

Algorithm 1 Stationary-Weighted Fitted Q -Evaluation

Input: Offline data $\mathcal{D}_n$, target policy π , function class $\mathcal{F}$, iterations K , discount γ

```

    # Estimate stationary state-action density ratio
    1:  $\hat{w}_\pi \leftarrow$  estimator of  $w_\pi$ 
    2: Initialize  $\hat{Q}^{(0)} \in \mathcal{F}$ 
    3: for  $k = 0, \dots, K - 1$  do
        # Construct Bellman targets
    4:  $Y_i^{(k)} \leftarrow R_i + \gamma(\pi\hat{Q}^{(k)})(S'_i)$ 
        # Stationary-weighted regression
    5:  $\hat{Q}^{(k+1)} \leftarrow \arg \min_{f \in \mathcal{F}} \sum_{i=1}^n \hat{w}_\pi(S_i, A_i) \{Y_i^{(k)} - f(S_i, A_i)\}^2$ 
    6: end for
Output:  $\hat{Q}^{(K)}$ 

```

B.2 Extension to Admissible Weighting Functions

For clarity, the main text focuses on the stationary state-action density ratio $d\mu_\pi/d\nu_b$. The more general analysis separates two requirements. The first is *projection equivalence*: the reference weight must reproduce the stationary projected Bellman update on $\mathcal{F}$. This condition is weaker than Bellman

648 Then Condition C7 holds if

$$\rho_G := \left\| G^{-1/2} \Delta_G G^{-1/2} \right\|_{\text{op}} < 1,$$

649 where the inverse is taken on the support of G . In this case, $\rho_{\mathcal{H}} \leq \rho_G$, and $\hat{G} := G + \Delta_G$ satisfies

$$(1 - \rho_G)u^\top G u \leq u^\top \hat{G} u \leq (1 + \rho_G)u^\top G u \quad \text{for all } u.$$

650 *Remark 1* (Localized curvature). The global curvature condition in Condition C7 can be weakened to
 651 a localized version. For $r > 0$, define

$$\rho_{\mathcal{H}}(r) := \sup_{\substack{h \in \mathcal{H}_{\mathcal{F}} \\ \|h\|_{2, w_{\pi} \nu_b} \geq r}} \left| \frac{E_{\nu_b}[(\hat{w}_{\pi} - w_{\pi})h^2]}{E_{\nu_b}[w_{\pi}h^2]} \right|.$$

652 If $\rho_{\mathcal{H}}(r) < 1$, the proof of Lemma 10 yields the one-step bound

$$\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_{2, w_{\pi} \nu_b} \leq \max \left\{ r, \frac{C}{1 - \rho_{\mathcal{H}}(r)} \left[\kappa_{\text{cov}} \left(\delta_n + \sqrt{\frac{\log(1/\tau)}{n}} \right) + \omega_{\text{Bell}, w} \right] \right\}.$$

653 Indeed, if the fitted error direction has weighted norm below r , the bound is immediate; otherwise,
 654 the same quadratic-curvature argument applies with $\rho_{\mathcal{H}}(r)$ in place of $\rho_{\mathcal{H}}$.

655 The useful choice is to take r at the one-step error scale. Writing

$$a_n := \kappa_{\text{cov}} \left(\delta_n + \sqrt{\frac{\log(1/\tau)}{n}} \right) + \omega_{\text{Bell}, w},$$

656 one may choose $r \asymp a_n$, or more generally the smallest radius for which $\rho_{\mathcal{H}}(r) \leq \rho_0 < 1$. Then the
 657 localized bound gives

$$\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_{2, w_{\pi} \nu_b} \lesssim a_n.$$

658 Thus curvature is only needed above the statistical error scale: if the fitted error is smaller than r ,
 659 the desired bound already holds, while larger directions are controlled by the localized curvature
 660 condition.

661 A sufficient localized condition is

$$B_{\mathcal{H}}(r) \left\| \frac{\hat{w}_{\pi}}{w_{\pi}} - 1 \right\|_{2, w_{\pi} \nu_b} < 1, \quad B_{\mathcal{H}}(r) := \sup_{\substack{h \in \mathcal{H}_{\mathcal{F}} \\ \|h\|_{2, w_{\pi} \nu_b} \geq r}} \frac{\|h\|_{4, w_{\pi} \nu_b}^2}{\|h\|_{2, w_{\pi} \nu_b}^2}.$$

662 This localized form is useful for rich classes in which the global L^4 -to- L^2 comparison may fail
 663 because of highly concentrated directions, while curvature above the statistical error scale remains
 664 controlled.

665 *Remark 2* (Approximate Bellman completeness). Approximate Bellman completeness enters through
 666 $\omega_{\text{Bell}, w}$. A simple sup-norm bound is

$$\omega_{\text{Bell}, w} \leq c \left\| \frac{\hat{w}_{\pi}}{w_{\pi}} - 1 \right\|_{\infty} \varepsilon_{\text{Bell}},$$

667 with the same extended-ratio convention on $\{w_{\pi} = 0\}$. More generally, if

$$\|\mathcal{T}Q - \mathcal{T}_{\mathcal{F}}Q\|_{\infty} \leq L_{\mathcal{R}} \|\mathcal{T}Q - \mathcal{T}_{\mathcal{F}}Q\|_{2, \mu_{\pi}}^{\beta} \quad \text{for all } Q \in \mathcal{F},$$

668 then

$$\omega_{\text{Bell}, w} \leq L_{\mathcal{R}} \varepsilon_w \varepsilon_{\text{Bell}}^{\beta}.$$

669 This residual interpolation condition holds with $\beta = 0$ under uniform boundedness of the Bellman
 670 residual class. Larger exponents are available when the residual class $\{\mathcal{T}Q - \mathcal{T}_{\mathcal{F}}Q : Q \in \mathcal{F}\}$ is
 671 smoother or lower-dimensional. For a finite-dimensional linear residual class with a nondegenerate
 672 Gram matrix and bounded features, norm equivalence gives $\beta = 1$. For a d -variate Hölder ball of
 673 smoothness $s > 0$, the standard Hölder interpolation inequality gives

$$\|r\|_{\infty} \leq C \|r\|_{2, \mu_{\pi}}^{2s/(2s+d)},$$

so $\beta = 2s/(2s + d)$. For Sobolev or RKHS balls of smoothness $s > d/2$, the Gagliardo–Nirenberg/Sobolev interpolation inequality gives

$$\|r\|_\infty \leq C \|r\|_{2, \mu_\pi}^{1-d/(2s)},$$

so $\beta = 1 - d/(2s)$. For signed convex hulls of uniformly bounded dictionaries, one generally obtains only the bounded-class case $\beta = 0$. If the hull is restricted to an m -dimensional or s_0 -sparse subspace with a nondegenerate Gram matrix, the finite-dimensional norm-equivalence case gives $\beta = 1$, with constants depending on m or s_0 . These are standard interpolation and entropy consequences for RKHS balls, signed convex hulls, and Hölder or Sobolev classes [1, 10, 51, 68, 74].

B.4 Discussion of Main Result

Contraction under approximate stationary density-ratio weights. The analysis extends beyond exact recovery of w_π . Suppose $\bar{w}_\pi$ is a limiting weight and the projected Bellman operator is contractive in $L^2(\bar{w}_\pi \nu_b)$ with factor $\bar{\gamma} < 1$. Then the same inexact Picard argument applies with $\bar{\gamma}$ in place of γ , and with the corresponding projected fixed point under the $\bar{w}_\pi \nu_b$ -weighted norm. Thus, even misspecified weights can improve stability if they move the projection norm closer to one in which the Bellman operator is contractive.

A simple sufficient condition is multiplicative closeness to the stationary ratio:

$$(1 - \varepsilon)w_\pi(s, a) \leq \bar{w}_\pi(s, a) \leq (1 + \varepsilon)w_\pi(s, a).$$

Then $L^2(\bar{w}_\pi \nu_b)$ and $L^2(w_\pi \nu_b) = L^2(\mu_\pi)$ are equivalent. In particular,

$$\|f\|_{2, \mu_\pi}^2 \leq \frac{1}{1 - \varepsilon} \|f\|_{2, \bar{w}_\pi \nu_b}^2, \quad \|f\|_{2, \bar{w}_\pi \nu_b}^2 \leq (1 + \varepsilon) \|f\|_{2, \mu_\pi}^2.$$

Since $\mathcal{T}$ is a γ -contraction in $L^2(\mu_\pi)$, it is a

$$\bar{\gamma} \leq \gamma \sqrt{\frac{1 + \varepsilon}{1 - \varepsilon}}$$

contraction in $L^2(\bar{w}_\pi \nu_b)$, provided the right-hand side is less than one. Thus small multiplicative weight errors preserve contraction up to an inflated contraction factor.

On-policy and ergodic limits. When $\nu_b = \mu_\pi$, the stationary state-action density ratio is $w_\pi \equiv 1$, so stationary-weighted FQE reduces to ordinary FQE and the weight-related terms vanish. More generally, under ergodic on-policy trajectory sampling, empirical state-action visitation frequencies converge to μ_π . Thus the empirical regression norm approaches the stationary norm, and ordinary FQE inherits the same contraction property asymptotically, up to sampling error.

B.5 Implicit Stationary Weighting by Sampling

Stationary weighting can also be achieved implicitly when the transition kernel P , and hence μ_π , is known. If we can sample $(\tilde{S}_i, \tilde{A}_i, \tilde{S}'_i) \sim \mu_\pi \otimes P$ directly, then the regression design is already matched to the target stationary distribution. The only missing quantity is the reward, which must be estimated from the behavior data. Let $\hat{r}$ denote such an estimate, and run unweighted FQE on

$$\{(\tilde{S}_i, \tilde{A}_i, \hat{r}(\tilde{S}_i, \tilde{A}_i), \tilde{S}'_i)\}_{i=1}^n.$$

This is ordinary stationary-sampling FQE with $w_\pi \equiv 1$, but with a perturbed reward.

Define

$$\hat{\mathcal{T}}Q := \hat{r} + \gamma P_\pi Q, \quad \hat{\mathcal{T}}_\mathcal{F} := \Pi_\mathcal{F} \hat{\mathcal{T}},$$

and let $\hat{Q}_\mathcal{F}^\pi$ be the fixed point of $\hat{\mathcal{T}}_\mathcal{F}$. For the K -step iterate $\hat{Q}^{(K)}$,

$$\|\hat{Q}^{(K)} - Q_\mathcal{F}^\pi\|_{2, \mu_\pi} \leq \|\hat{Q}^{(K)} - \hat{Q}_\mathcal{F}^\pi\|_{2, \mu_\pi} + \|\hat{Q}_\mathcal{F}^\pi - Q_\mathcal{F}^\pi\|_{2, \mu_\pi}.$$

The first term is the usual finite-iteration and statistical error under stationary sampling. The second term is the reward-estimation perturbation.

708 **Theorem 3** (Reward perturbation). *Under Conditions C1–C2,*

$$\|\widehat{Q}_{\mathcal{F}}^{\pi} - Q_{\mathcal{F}}^{\pi}\|_{2,\mu_{\pi}} \leq \frac{1}{1-\gamma} \|\widehat{r} - r_0\|_{2,\mu_{\pi}}.$$

709 *If, in addition, $\mathcal{F}$ is a linear subspace of $L^2(\mu_{\pi})$, then*

$$\|\widehat{Q}_{\mathcal{F}}^{\pi} - Q_{\mathcal{F}}^{\pi}\|_{2,\mu_{\pi}} \leq \frac{1}{1-\gamma} \|\Pi_{\mathcal{F}}(\widehat{r} - r_0)\|_{2,\mu_{\pi}}.$$

710 C Details on Stationary-Ratio Estimation

711 Stationary-weighted FQE requires weights approximating the stationary state-action density ratio
 712 $w_{\pi} = d\mu_{\pi}/d\nu_b$. Estimating such ratios is a well-studied problem, often formulated through DICE-
 713 style saddle-point objectives [43, 56, 92] or related balancing-weight methods [70, 83]. Stationary
 714 ratios are the undiscounted analogues of discounted occupancy ratios and can be obtained as a $\gamma \rightarrow 1$
 715 limit under suitable normalization and standard ergodicity conditions.

716 The key identifying condition is stationarity. Since $w_{\pi}\nu_b = \mu_{\pi}$ is stationary for (π, P) , the ratio
 717 satisfies

$$E[w_{\pi}(S, A)\{g(S, A) - g(S', A')\}] = 0 \quad \text{for all } g, \quad (2)$$

718 where $(S, A) \sim \nu_b$, $S' \sim P(\cdot | S, A)$, and $A' \sim \pi(\cdot | S')$. Conversely, if a nonnegative function w
 719 satisfies (2) for all bounded measurable g and $E_{\nu_b} w(S, A) = 1$, then $w\nu_b$ is a stationary state-action
 720 distribution for (π, P) . Thus, when the stationary distribution is unique, $w = w_{\pi}$ ν_b -almost surely.

721 Minimax and DICE-style estimators enforce (2) over a critic class $\mathcal{G}$ while searching over a ratio
 722 class $\mathcal{W}$. A representative empirical estimator is

$$\begin{aligned} \widehat{w}_{\pi} \in \arg \min_{w \in \mathcal{W}} \left[\sup_{g \in \mathcal{G}} \left\{ \frac{1}{n} \sum_{i=1}^n w(S_i, A_i) \{g(S_i, A_i) - g(S'_i, A'_i)\} - \frac{1}{2n} \sum_{i=1}^n g(S_i, A_i)^2 \right\} \right. \\ \left. + \left(\frac{1}{n} \sum_{i=1}^n w(S_i, A_i) - 1 \right)^2 \right], \end{aligned} \quad (3)$$

723 where $A'_i \sim \pi(\cdot | S'_i)$. The final term enforces normalization. The quadratic term in g makes the inner
 724 problem strongly concave and turns the supremum into a squared measure of stationarity-moment
 725 violation. Other DICE variants use different convex regularizers, discounted-moment constraints,
 726 or additional normalization and positivity penalties, but they share the same stationarity-moment
 727 structure.

728 When $\mathcal{G}$ is linear or an RKHS, the inner maximization in (3) often has a closed form. For example,
 729 suppose $g_{\theta}(x) = \theta^{\top} \phi(x)$, where $x = (s, a)$. For a fixed candidate ratio w , define

$$\widehat{m}_w := \frac{1}{n} \sum_{i=1}^n w(S_i, A_i) \{\phi(S_i, A_i) - \phi(S'_i, A'_i)\}, \quad \widehat{\Sigma} := \frac{1}{n} \sum_{i=1}^n \phi(S_i, A_i) \phi(S_i, A_i)^{\top}.$$

730 Adding ridge regularization $\lambda \|\theta\|_2^2/2$, the inner supremum is

$$\sup_{\theta} \left\{ \theta^{\top} \widehat{m}_w - \frac{1}{2} \theta^{\top} (\widehat{\Sigma} + \lambda I) \theta \right\} = \frac{1}{2} \widehat{m}_w^{\top} (\widehat{\Sigma} + \lambda I)^{-1} \widehat{m}_w.$$

731 Thus the ratio estimator can be viewed as choosing w to balance stationary flow moments in the
 732 feature space ϕ . RKHS critics give an analogous kernelized discrepancy between the weighted current
 733 state-action distribution and the induced next state-action distribution. These structured critic classes
 734 can yield stable, computationally simple weights while still approximating the stationary projection
 735 norm [17, 59, 70, 83].

736 For numerical stability, practical implementations often regularize or truncate the estimated ratio,
 737 for example using Tikhonov regularization [72]. Common choices include constraining $w \geq 0$,
 738 adding an ℓ_2 or entropy penalty on w , clipping large weights, or estimating a discounted occupancy
 739 ratio with discount close to one. Such regularization trades small bias for lower variance and better
 740 conditioning, which is often important under limited overlap.

741 **Instantiating relative weight error.** The generic term $\|\widehat{w}_\pi/w_\pi - 1\|_{2,w_\pi\nu_b}$ is useful for sufficient
 742 bounds on the residual interaction and can be controlled using guarantees for specific minimax or
 743 saddle-point ratio estimators. Analyses for conditional moment problems are given by Dikkala et al.
 744 [17], Olivas-Martinez and Rotnitzky [59], and off-policy weighting analyses are given by Uehara
 745 et al. [70], Wang et al. [83]. At a high level, such bounds decompose into weight-class approximation,
 746 critic-class approximation, and statistical error:

$$\inf_{w \in \mathcal{W}} \|w - w^\dagger\| + \inf_{g \in \mathcal{G}} \|g - g^\dagger\| + \text{stat}(\mathcal{W}, \mathcal{G}, n),$$

747 where $\mathcal{W}$ and $\mathcal{G}$ are the ratio and critic classes, and $\text{stat}(\mathcal{W}, \mathcal{G}, n)$ is controlled by localized complex-
 748 ity; see also Bennett et al. [8].

749 C.1 Fixed-Point Estimation of Stationary Ratios

750 The stationary state-action density ratio $w_\pi = d\mu_\pi/d\nu_b$ is identified by the fixed-point equation
 751 implied by stationarity $\mu_\pi = \mu_\pi P_\pi$:

$$w_\pi(s', a') = (\mathcal{K}w_\pi)(s', a'),$$

752 where the linear operator $\mathcal{K}$ is

$$(\mathcal{K}g)(s', a') = \frac{\pi(a' | s')}{\nu_b(a' | s')} \mathbb{E}_{(S,A) \sim \nu_b} [g(S, A) k(S, A, s')],$$

753 and

$$k(s, a, s') = \frac{P(s' | s, a)}{\rho_{b,1}(s')}, \quad \rho_{b,1}(s') := \mathbb{E}_{(S,A) \sim \nu_b} [P(s' | S, A)].$$

754 Here k is the transition density ratio and $\rho_{b,1}$ is the marginal density of S' induced by the behavior
 755 dynamics. The quantity k can be estimated directly using the empirical squared-loss objective of
 756 Hines and Miles [32]:

$$\frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n k(S_i, A_i, S'_j)^2 - \frac{2}{n} \sum_{i=1}^n k(S_i, A_i, S'_i).$$

757 However, $\mathcal{K}$ is the adjoint of P_π in $L^2(\mu_\pi)$ and satisfies $\|\mathcal{K}\|_{L^2(\mu_\pi)} = 1$, so direct Picard iteration of
 758 $w_\pi = \mathcal{K}w_\pi$ is not contractive. A standard workaround is the discounted resolvent equation

$$d_{\gamma'} = (1 - \gamma') \frac{\pi}{\nu_b} + \gamma' \mathcal{K}d_{\gamma'},$$

759 whose solution $d_{\gamma'}$ is the γ' -discounted occupancy ratio. The resolvent replaces exact stationarity by
 760 a γ' -discounted moment balance, which yields a γ' -contractive update. Since $\mathcal{K}$ is nonexpansive in
 761 $L^2(\mu_\pi)$, Picard iteration applies and $d_{\gamma'}$ can be computed analogously to fitted value iteration.

762 Under standard ergodicity assumptions—irreducibility, aperiodicity, and geometric mixing—the
 763 discounted ratios satisfy $d_{\gamma'} \rightarrow w_\pi$ as $\gamma' \uparrow 1$ [28, 53]. The resolvent formulation induces a familiar
 764 bias–variance trade-off: larger $1 - \gamma'$ improves numerical stability but introduces bias of order
 765 $O(1 - \gamma')$, whereas letting $\gamma' \uparrow 1$ removes the bias at the expense of variance scaling like $(1 - \gamma')^{-1}$
 766 due to the effective horizon of the discounted occupancy measure.

767 D Experiments

768 D.1 Finite-State Diagnostic Experiments

769 **Baird hub-and-spokes.** The Baird experiment uses a hub state and six spoke states [5]. The solid
 770 action moves deterministically to the hub; the dashed action moves uniformly to one of the spokes.
 771 The target policy always takes the solid action. The behavior policy takes the solid action with
 772 probability κ , so decreasing κ reduces stationary overlap with the target policy. We use the standard
 773 seven-dimensional linear feature map, zero rewards, ridge-regularized least squares at each FQE step,
 774 and exact stationary ratios. Since $Q^\pi \equiv 0$, the experiment isolates whether the projected Bellman
 775 iteration contracts in the intended norm. Reported curves are averaged over 100 independent datasets.

776 **Sparse Garnet MDPs.** The Garnet experiment uses random MDPs with 100 states, 4 actions,
777 and branching factor 5. For each state-action pair, five next states are sampled without replacement
778 and assigned Dirichlet transition probabilities. Rewards are Gaussian. The target policy is sampled
779 randomly; the behavior policy is ε -greedy with $\varepsilon = 0.1$. Data are generated with a reset scheme
780 that deliberately breaks stationarity. The linear feature class has dimension 5 and contains Q^π , but
781 is generically not Bellman complete. Thus any instability is due to the fitted projection geometry,
782 not ordinary realizability failure. The main figure uses exact stationary ratios; the same qualitative
783 stabilization was also observed with tabular DualDICE-style ratio estimates.

784 D.2 Controlled linear-Gaussian simulation

785 The continuous benchmark uses $S_t \in \mathbb{R}^2$ and $A_t \in \mathbb{R}$. The dynamics are linear Gaussian,

$$S_{t+1} = BS_t + CA_t + \epsilon_t, \quad \epsilon_t \sim N(0, \sigma^2 I),$$

786 with quadratic reward

$$r(s, a) = -\|s - s^*\|^2 - \lambda a^2.$$

787 The target and behavior policies are Gaussian linear policies,

$$A_t^\pi | S_t = s \sim N(K_\pi s, \sigma_\pi^2), \quad A_t^b | S_t = s \sim N(K_b s, \sigma_b^2),$$

788 where $K_b = K_\pi + \Delta v$. The scalar Δ controls the behavior–target shift. We evaluate 100 independent
789 replications over

$$\Delta \in \{0, 0.25, 0.5, 0.75, 0.9, 1.0, 1.1, 1.2, 1.25, 1.35, \\ 1.5, 1.6, 1.7, 1.8, 1.9, 1.95, 2.0, 2.025, 2.05, 2.055, 2.06\}.$$

790 The shift direction is scaled so that the closed-loop behavior process remains stable throughout the
791 sweep. The target action standard deviation is 0.10, the behavior action standard deviation is 0.15,
792 and the process-noise standard deviation is 0.12. Thus $\Delta = 0$ gives high overlap, but is not exactly
793 on-policy because the behavior and target policies have different action variances.

794 Because the system is linear Gaussian with quadratic rewards, Q^π , V^π , and the discounted policy
795 value ψ_π are computed analytically by a quadratic fixed-point recursion. For the main stationary
796 weighting experiments, the reference distribution is the stationary Gaussian state-action distribution
797 under the target policy, and oracle weights are stationary Gaussian density ratios. Discounted
798 occupancy variants are evaluated using Gaussian-mixture representations over time.

799 Offline data are sampled from the behavior stationary distribution. The value discount in the Bellman
800 targets is $\gamma = 0.95$. The regression weights target the stationary state-action ratio $d\mu_\pi/d\nu_b$, where
801 μ_π is the target stationary state-action distribution and ν_b is the behavior sampling distribution.

802 We use the deliberately misspecified affine class $[1, s_1, s_2, a]$. Linear FQE uses analytic target-action
803 expectations, so the Bellman targets do not include Monte Carlo integration error.

804 The ratio estimator is the finite-dimensional closed-form version of the stationarity minimax moment
805 problem, with a mean-one normalization. Let $z = (s, a)$. The ratio features $\phi_w(z)$ consist of
806 $(1, z_1, z_2, z_3, z_1^2, z_2^2, z_3^2)$ and 64 Gaussian RBF features on z ; the centers are evenly spaced behavior-
807 sample points and the bandwidth is the median center distance. Define

$$\Delta_\pi \phi_w(s, a, s') = \phi_w(s, a) - \gamma_{\text{ratio}} E_{a' \sim \pi(\cdot | s')} \{\phi_w(s', a')\}, \quad m_0 = E_{s_0, a_0 \sim \pi} \{\phi_w(s_0, a_0)\}.$$

808 For $w_\alpha(z) = \phi_w(z)^\top \alpha$, the reduced minimax estimator solves

$$\min_\alpha \|\mathbb{P}_n w_\alpha(Z) \Delta_\pi \phi_w - (1 - \gamma_{\text{ratio}}) m_0\|_{(B_n + \lambda_d I)^{-1}}^2 + \lambda_p \|\alpha\|_2^2 + \lambda_0 \{\mathbb{P}_n w_\alpha(Z) - 1\}^2,$$

809 where $B_n = \mathbb{P}_n \phi_w(Z) \phi_w(Z)^\top$, $\lambda_0 = 10$, and $\gamma_{\text{ratio}} = 1$ unless otherwise stated. In the stationary
810 case, the flow right-hand side is zero and the mean-one penalty fixes the scale. Negative fitted weights
811 are floored at 10^{-8} , and all weights are normalized to have empirical mean one before fitting FQE.

812 We use 64 standardized RBF ratio features and also include a positive exponential-quadratic moment
813 model

$$w_\theta(s, a) = \exp\{\theta^\top \psi(s, a)\},$$

814 where ψ contains standardized quadratic state-action features. Both estimated ratio classes are fit only
815 from stationarity moments. For stability, we consider several additional regularization and truncation

choices. Fixed truncation uses $\bar{w}_i \propto \min\{w_i, 2\}$. ESS-adaptive winsorization chooses the largest empirical cap c such that

$$\text{ESS}(\bar{w})/n = \frac{(\sum_i \bar{w}_i)^2}{n \sum_i \bar{w}_i^2} \geq 0.60.$$

CV truncation chooses $c \in \{1.5, 2, 3, 5, 10, 25, \infty\}$ by three-fold held-out stationarity risk, evaluating the ratio objective on held-out (S, A, S') after fitting α on the other folds. CV-Tikhonov chooses $\lambda_p = \lambda_d$ from

$$\{10^{-5}, 3 \cdot 10^{-5}, 10^{-4}, 3 \cdot 10^{-4}, \dots, 3, 10\}$$

by five-fold held-out ratio risk and then refits on all offline data. These selection criteria use only transition tuples (S, A, S') , not rewards or oracle evaluation targets.

The primary metric is mean squared Q -error under the target stationary distribution. Effective sample size, upper weight quantiles, and maximum weights are reported as overlap diagnostics as the behavior policy moves away from the target policy.

We also include a direct minimax Bellman-residual baseline. With $Q_\theta(s, a) = \phi_Q(s, a)^\top \theta$ and critic features $\psi(s, a)$, this baseline solves

$$\min_{\theta} \|\mathbb{P}_n \psi(S, A) \{Q_\theta(S, A) - R - \gamma E_\pi Q_\theta(S', A')\}\|_{(\mathbb{P}_n \psi \psi^\top + \lambda_d I)^{-1}}^2 + \lambda_q \|\theta\|_2^2.$$

The main comparison uses the same affine Q -class as FQE and the same RBF critic features used for ratio fitting. The practical minimax row chooses a dimensionless Tikhonov level η by held-out Bellman moment risk, using

$$\lambda_d = \eta \text{tr}(\mathbb{P}_n \psi \psi^\top) / d_\psi$$

and the analogous normalized ridge for the reduced Q -system. This baseline is not a weighted FQE estimator; it is included to compare stationary weighting with a standard Bellman-moment stabilization. For local RBF weighting, correctly specified quadratic weighting, and minimax fitting, we also report oracle-tuned diagnostics that select the regularization level by target-stationary Q -error on an independent evaluation sample. These oracle-tuned rows are diagnostic and are not used as deployable tuning rules.

D.3 Practical and oracle-tuned Bellman baseline comparison

Figure 3 compares deployable and oracle-tuned versions of the stationary-weighted and minimax Bellman baselines. The results show that minimax Bellman methods can perform very well when tuned to oracle target-stationary error, consistent with their theoretical motivation, but also illustrate a common practical challenge for minimax approaches: deployable tuning by held-out moment risk can be difficult and need not select the configuration with the best target-stationary Q -error.

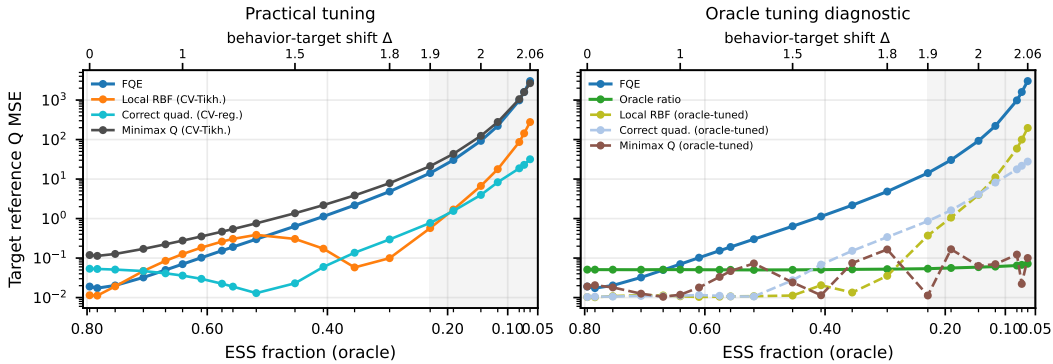


Figure 3: **Practical and oracle-tuned Bellman baselines.** Both panels plot target-stationary Q -error versus oracle ESS fraction, with behavior–target shift shown on the top axis. Left: deployable tuning rules based on held-out moment risks. Right: oracle-tuned diagnostics that select ridge parameters using independent target-stationary error. Practical stationary weighting improves over unweighted FQE in lower-overlap regimes, while oracle-tuned weighting and minimax rows show the attainable performance of the corresponding estimator classes.

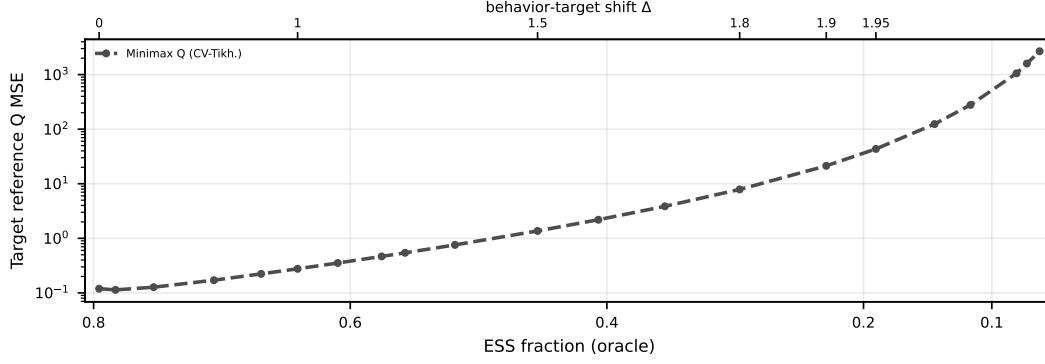


Figure 4: **Minimax regularization sensitivity.** The minimax Bellman baseline is sensitive to the normalized Tikhonov level: weak regularization is high variance, while overly large ridges bias the Bellman moment fit. The CV-Tikhonov row selects the ridge by held-out Bellman moment risk.

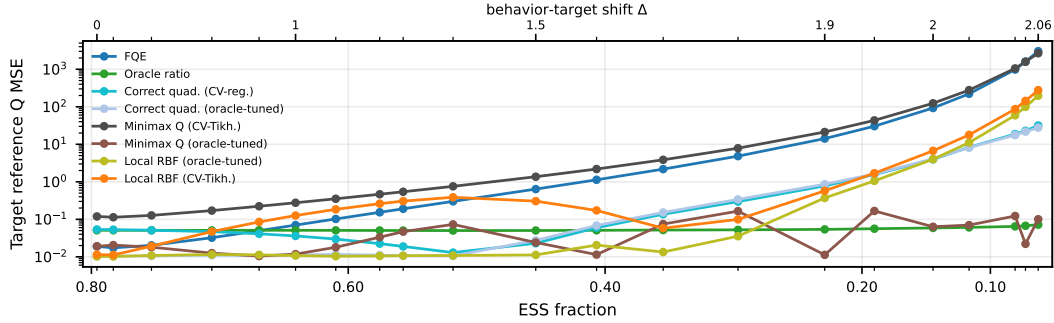


Figure 5: **Stabilization comparison in the controlled linear-Gaussian benchmark.** Standard FQE is fit in the behavior norm. Oracle stationary weighting shows the target-stationary projection mechanism. The estimated-weight rows compare unregularized weights, a prespecified cap, ESS-adaptive winsorization, CV-Tikhonov ratio regularization, and CV truncation. The CV criteria use held-out stationarity moment risk and do not use rewards or oracle evaluation targets.

843 As a diagnostic control, we also ran a flexible outer Q -class using the same standardized RBF feature
 844 map as the ratio critic. In this setting, standard FQE is already accurate under good overlap, while
 845 minimax fitting is stable but more biased; under the most severe shifts, minimax remains finite when
 846 ordinary FQE extrapolates poorly. This check confirms that the main linear-Gaussian benchmark is
 847 not driven by a degenerate simulator, but it is not used as the primary comparison.

848 E Contraction Results

849 E.1 Contraction Under Stationary Measures

850 **Lemma 4** (Bellman contraction for Q -functions under a stationary measure). *If μ_π is stationary for*
 851 *P_π , then P_π is nonexpansive in $L^2(\mu_\pi)$:*

$$\|P_\pi(Q_1 - Q_2)\|_{2, \mu_\pi} \leq \|Q_1 - Q_2\|_{2, \mu_\pi}.$$

852 *Therefore the Bellman operator $\mathcal{T}$ is a γ -contraction:*

$$\|\mathcal{T}Q_1 - \mathcal{T}Q_2\|_{2, \mu_\pi} \leq \gamma \|Q_1 - Q_2\|_{2, \mu_\pi}.$$

853 *Proof.* Let $h := Q_1 - Q_2$. For any (s, a) , Jensen's inequality gives

$$(P_\pi h)(s, a)^2 \leq (P_\pi(h^2))(s, a),$$

854 since P_π is a conditional expectation operator.

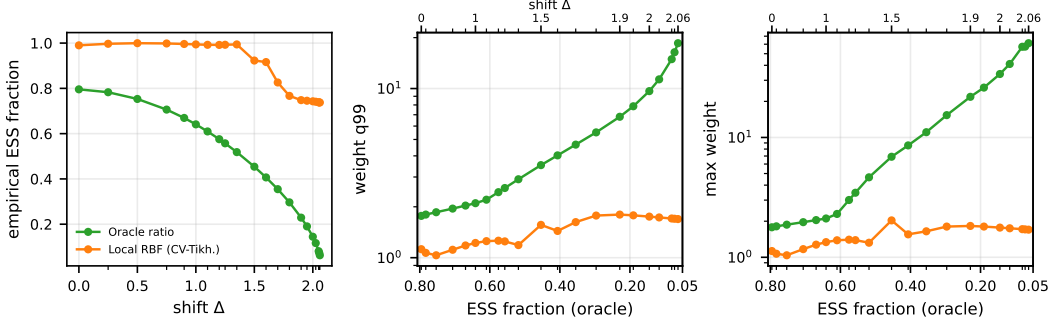


Figure 6: **Weight diagnostics for all stabilizers in the controlled linear-Gaussian benchmark.** The main text reports only the oracle, unregularized, and CV-Tikhonov weights. Here we also show the prespecified cap, ESS-adaptive winsorization, and CV-truncated cap used in the stabilizer comparison.

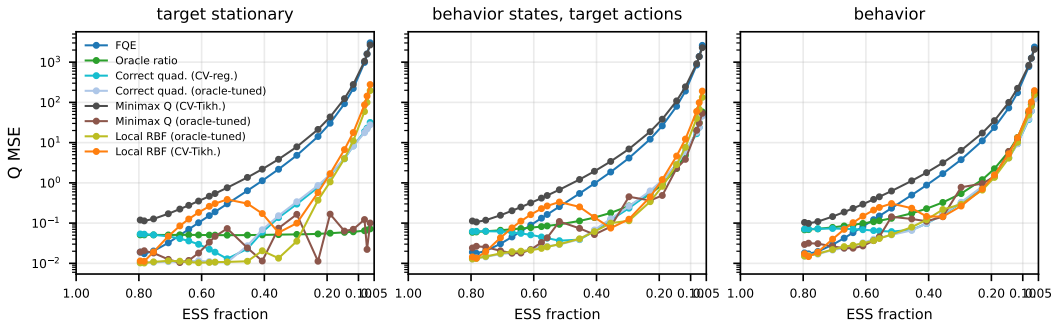


Figure 7: **Norm diagnostics for the controlled linear-Gaussian benchmark.** Target-stationary error is the policy-evaluation norm used in the main panel. The behavior-state/target-action and behavior-norm panels show that the gains are not an artifact of evaluating only one favorable distribution; oracle raw weights intentionally optimize the target stationary projection and become poor under behavior-weighted diagnostics when overlap is severe.

855 Integrating with respect to μ_π and using stationarity $\mu_\pi = \mu_\pi(P_\pi)$,

$$\|P_\pi h\|_{2,\mu_\pi}^2 = \int (P_\pi h)^2 d\mu_\pi \leq \int P_\pi(h^2) d\mu_\pi = \int h^2 d\mu_\pi.$$

856 Thus P_π is nonexpansive: $\|P_\pi h\|_{2,\mu_\pi} \leq \|h\|_{2,\mu_\pi}$.

857 For the Bellman operator,

$$\mathcal{T}Q_1 - \mathcal{T}Q_2 = \gamma P_\pi(Q_1 - Q_2) = \gamma P_\pi h.$$

858 Taking norms and applying nonexpansiveness proves

$$\|\mathcal{T}Q_1 - \mathcal{T}Q_2\|_{2,\mu_\pi} \leq \gamma \|h\|_{2,\mu_\pi},$$

859 as claimed. $\square$

860 **Lemma 5** (Contraction of the projected Bellman operator). *If μ_π is stationary for P_π , then $\mathcal{T}_\mathcal{F}$ is a*
861 *γ -contraction on $L^2(\mu_\pi)$; that is, for all $Q_1, Q_2 \in L^2(\mu_\pi)$*

$$\|\mathcal{T}_\mathcal{F}Q_1 - \mathcal{T}_\mathcal{F}Q_2\|_{2,\mu_\pi} \leq \gamma \|Q_1 - Q_2\|_{2,\mu_\pi}.$$

862 *Proof.* By Lemma 4,

$$\|\mathcal{T}Q_1 - \mathcal{T}Q_2\|_{2,\mu_\pi} \leq \gamma \|Q_1 - Q_2\|_{2,\mu_\pi}.$$

863 Since $\Pi_\mathcal{F}$ is the $L^2(\mu_\pi)$ orthogonal projection onto a closed convex set, it is nonexpansive:

$$\|\Pi_\mathcal{F}f - \Pi_\mathcal{F}g\|_{2,\mu_\pi} \leq \|f - g\|_{2,\mu_\pi}.$$

864 Thus

$$\begin{aligned}\|\mathcal{T}_{\mathcal{F}}Q_1 - \mathcal{T}_{\mathcal{F}}Q_2\|_{2,\mu_\pi} &= \|\Pi_{\mathcal{F}}(\mathcal{T}Q_1) - \Pi_{\mathcal{F}}(\mathcal{T}Q_2)\|_{2,\mu_\pi} \\ &\leq \|\mathcal{T}Q_1 - \mathcal{T}Q_2\|_{2,\mu_\pi} \\ &\leq \gamma \|Q_1 - Q_2\|_{2,\mu_\pi}.\end{aligned}$$

865

□

866 *Proof of Lemma 1.* The first claim follows directly from Lemma 4. The second claim follows from
867 Lemma 5. □

868 E.2 Extension to Other Weighting Functions

869 Our analysis extends beyond the stationary ratio $w_\pi = d\mu_\pi/d\nu_b$. More generally, the same arguments
870 apply to any weighting measure under which the Bellman operator $\mathcal{T}$ is a $\bar{\gamma}$ -contraction in L^2 , where
871 $\bar{\gamma}$ may differ from the environment discount factor γ . The next result shows that β -discounted
872 occupancy weighting gives one such example, with contraction modulus $\gamma/\sqrt{\beta}$. This is especially
873 useful because discounted occupancy ratios with $\beta < 1$ can be defined without the ergodicity
874 conditions required for stationary ratios.

875 **Theorem 4** (Contraction under discounted-occupancy weighting). *Fix a policy π and $\gamma \in (0, 1)$. For*
876 *$\beta \in (0, 1]$, let*

$$w_\beta := (1 - \beta) \sum_{t=0}^{\infty} \beta^t \rho_t,$$

877 *where ρ_t denotes the distribution of (S_t, A_t) under initial state distribution ρ and policy π . Equiva-*
878 *lently,*

$$w_\beta = (1 - \beta)(\rho \otimes \pi) + \beta w_\beta P_\pi.$$

879 *Then, for any measurable Q_1, Q_2 ,*

$$\|\mathcal{T}Q_1 - \mathcal{T}Q_2\|_{L^2(w_\beta)} \leq \frac{\gamma}{\sqrt{\beta}} \|Q_1 - Q_2\|_{L^2(w_\beta)}.$$

880 *Consequently, $\mathcal{T}$ is a contraction on $L^2(w_\beta)$ whenever*

$$\frac{\gamma}{\sqrt{\beta}} < 1, \quad \text{equivalently,} \quad \beta > \gamma^2.$$

881 *In particular, if $\beta = \gamma$, then the contraction modulus is $\sqrt{\gamma}$, and if $\beta = 1$, the modulus is γ .*

882 *Proof.* Let $\Delta := Q_1 - Q_2$. Since the reward term cancels,

$$\mathcal{T}Q_1 - \mathcal{T}Q_2 = \gamma P_\pi \Delta,$$

883 so it suffices to bound $\|P_\pi \Delta\|_{L^2(w_\beta)}$.

884 By conditional Jensen,

$$|(P_\pi \Delta)(s, a)|^2 = |\mathbb{E}[\Delta(S', A') \mid S = s, A = a]|^2 \leq \mathbb{E}[\Delta(S', A')^2 \mid S = s, A = a].$$

885 Integrating with respect to w_β gives

$$\begin{aligned}\|P_\pi \Delta\|_{L^2(w_\beta)}^2 &= \int |(P_\pi \Delta)(s, a)|^2 dw_\beta(s, a) \\ &\leq \int \mathbb{E}[\Delta(S', A')^2 \mid S = s, A = a] dw_\beta(s, a) \\ &= \int \Delta(s', a')^2 d(w_\beta P_\pi)(s', a').\end{aligned}$$

886 Hence

$$\|P_\pi \Delta\|_{L^2(w_\beta)}^2 \leq \|\Delta\|_{L^2(w_\beta P_\pi)}^2.$$

887 Next, the identity

$$w_\beta = (1 - \beta)(\rho \otimes \pi) + \beta w_\beta P_\pi$$

888 implies

$$\beta w_\beta P_\pi \leq w_\beta.$$

889 Therefore $w_\beta P_\pi \ll w_\beta$, with Radon–Nikodym derivative bounded by $1/\beta$:

$$\frac{d(w_\beta P_\pi)}{dw_\beta} \leq \frac{1}{\beta}, \quad w_\beta\text{-a.e.}$$

890 It follows that

$$\begin{aligned} \|\Delta\|_{L^2(w_\beta P_\pi)}^2 &= \int \Delta(s, a)^2 \frac{d(w_\beta P_\pi)}{dw_\beta}(s, a) dw_\beta(s, a) \\ &\leq \frac{1}{\beta} \int \Delta(s, a)^2 dw_\beta(s, a) \\ &= \frac{1}{\beta} \|\Delta\|_{L^2(w_\beta)}^2. \end{aligned}$$

891 Combining the last two displays yields

$$\|P_\pi \Delta\|_{L^2(w_\beta)}^2 \leq \frac{1}{\beta} \|\Delta\|_{L^2(w_\beta)}^2,$$

892 and therefore

$$\|P_\pi \Delta\|_{L^2(w_\beta)} \leq \frac{1}{\sqrt{\beta}} \|\Delta\|_{L^2(w_\beta)}.$$

893 Multiplying by γ , we obtain

$$\|\mathcal{T}Q_1 - \mathcal{T}Q_2\|_{L^2(w_\beta)} = \gamma \|P_\pi \Delta\|_{L^2(w_\beta)} \leq \frac{\gamma}{\sqrt{\beta}} \|\Delta\|_{L^2(w_\beta)}.$$

894 This proves the bound. The contraction claim follows immediately when $\gamma/\sqrt{\beta} < 1$, equivalently
895 $\beta > \gamma^2$. $\square$

896 F Technical Lemmas

897 F.1 Local maximal inequality

898 Let $O_1, \dots, O_n \in \mathcal{O}$ be independent random variables. For any function $f : \mathcal{O} \rightarrow \mathbb{R}$, define

$$\|f\| := \sqrt{\frac{1}{n} \sum_{i=1}^n \mathbb{E}[f(O_i)^2]}. \quad (4)$$

899 For a star-shaped class of functions $\mathcal{F}$ and a radius $\delta \in (0, \infty)$, define the localized Rademacher
900 complexity

$$\mathcal{R}_n(\mathcal{F}, \delta) := \mathbb{E} \left[\sup_{\substack{f \in \mathcal{F} \\ \|f\| \leq \delta}} \frac{1}{n} \sum_{i=1}^n \epsilon_i f(O_i) \right],$$

901 where ϵ_i are i.i.d. Rademacher random variables.

902 The following lemma provides a local maximal inequality and restates Lemma 11 of [24] (see also
903 Lemma 11 of van der Laan et al. [77]).

904 **Lemma 6** (Local maximal inequality). *Let $\mathcal{F}$ be a star-shaped class of functions satisfying*
905 *$\sup_{f \in \mathcal{F}} \|f\|_\infty \leq M$. Let $\delta = \delta_n \in (0, 1)$ satisfy the critical radius condition $\mathcal{R}_n(\mathcal{F}, \delta) \leq \delta^2$,*
906 *and suppose that, as $n \rightarrow \infty$,*

$$\frac{1}{\sqrt{n}} \sqrt{\log \log(1/\delta_n)} = o(\delta_n).$$

907 Then there exists a constant $C > 0$ such that, for all $\tau \in (0, 1)$, with probability at least $1 - \tau$, every
 908 $f \in \mathcal{F}$ satisfies

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n (f(O_i) - \mathbb{E}[f(O_i)]) &\leq C \left(\delta^2 + \delta \|f\| \right) \\ &+ C \left(\frac{\sqrt{\log(1/\tau)} \|f\|}{\sqrt{n}} + \frac{M \log(1/\tau)}{n} \right). \end{aligned}$$

909 *Proof.* Lemma 11 in [77] shows that there exists a constant $C > 0$ (possibly depending on the choice
 910 of sequence $\{\delta_n\}$) such that, for all $u \geq 1$, with probability at least $1 - e^{-u^2}$, every $f \in \mathcal{F}$ satisfies

$$\frac{1}{n} \sum_{i=1}^n (f(O_i) - \mathbb{E}[f(O_i)]) \leq C \left(\delta_n^2 + \delta_n \|f\| + \frac{u \|f\|}{\sqrt{n}} + \frac{M u^2}{n} \right).$$

911 Set $u := \sqrt{\log(1/\tau)}$, so that $e^{-u^2} = e^{-\log(1/\tau)} = \tau$. Substituting this choice of u into the above
 912 inequality yields, with probability at least $1 - \tau$, every $f \in \mathcal{F}$ satisfies

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n (f(O_i) - \mathbb{E}[f(O_i)]) &\leq C \left(\delta_n^2 + \delta_n \|f\| \right) \\ &+ C \left(\frac{\sqrt{\log(1/\tau)} \|f\|}{\sqrt{n}} + \frac{M \log(1/\tau)}{n} \right). \end{aligned}$$

913 □

914 The following lemma bounds the localized Rademacher complexity in terms of the uniform entropy
 915 integral and is a direct consequence of Theorem 2.1 of Van Der Vaart and Wellner [78].

916 For any distribution Q and any uniformly bounded function class $\mathcal{F}$, let $N(\varepsilon, \mathcal{F}, L^2(Q))$ denote the
 917 ε -covering number of $\mathcal{F}$ under the $L^2(Q)$ norm [79]. Define the uniform entropy integral of $\mathcal{F}$ by

$$\mathcal{J}(\delta, \mathcal{F}) := \int_0^\delta \sup_Q \sqrt{\log N(\epsilon, \mathcal{F}, L^2(Q))} d\epsilon,$$

918 where the supremum is taken over all discrete probability distributions Q .

919 **Lemma 7.** Let $\mathcal{F}$ be a star-shaped class of functions such that $\sup_{f \in \mathcal{F}} \|f\|_\infty \leq M$. Then, for every
 920 $\delta > 0$,

$$\mathcal{R}_n(\mathcal{F}, \delta) \lesssim \frac{1}{\sqrt{n}} \mathcal{J}(\delta, \mathcal{F}) \left(1 + \frac{\mathcal{J}(\delta, \mathcal{F})}{\delta \sqrt{n}} \right),$$

921 where the implicit constant depends only on M .

922 *Proof.* This bound follows directly from the argument in the proof of Theorem 2.1 of Van Der Vaart
 923 and Wellner [78]; see in particular the step where the local Rademacher complexity is controlled by
 924 the uniform entropy integral for star-shaped classes. □

925 F.2 Regret of a Single Inexact Picard Step

926 Recall the weight function w_π in Condition C8, and let $\hat{w}_\pi$ denote an estimate of this weight that is
 927 trained independently of $\mathcal{D}_n$ (Condition C5).

928 Let $\mathcal{F}$ be a convex function class and let $\hat{Q}^{\text{init}} \in \mathcal{F}$ be an initial (possibly data-dependent) estimate.
 929 Define the weighted population risk

$$\hat{R}_0(Q) = \mathbb{E}_{\nu_b} \left[w_\pi(S, A) \{ R + \gamma(\pi \hat{Q}^{\text{init}})(S') - Q(S, A) \}^2 \right].$$

930 Following Algorithm 1, define the empirical minimizer

$$\hat{Q}_n := \arg \min_{Q \in \mathcal{F}} \hat{R}_n(Q),$$

931 where

$$\hat{R}_n(Q) = \sum_{i=1}^n \hat{w}_\pi(S_i, A_i) \{R_i + \gamma(\pi \hat{Q}^{\text{init}})(S'_i) - Q(S_i, A_i)\}^2.$$

932 Lemma 8 shows that this weighted population objective is correctly specified for the projected
933 Bellman operator $\mathcal{T}_\mathcal{F}(\hat{Q}^{\text{init}})$ under the projection equivalence in Condition C8.

934 **Lemma 8** (Fixed-point operator as population risk minimizer). *Under Conditions C8,*

$$\mathcal{T}_\mathcal{F}(\hat{Q}^{\text{init}}) = \arg \min_{Q \in \mathcal{F}} \hat{R}_0(Q).$$

935 *Proof.* By Condition C8 with $Q = \hat{Q}^{\text{init}}$, $\mathcal{T}_\mathcal{F}(\hat{Q}^{\text{init}})$ minimizes the $w_\pi \nu_b$ -weighted squared distance
936 to $\mathcal{T}(\hat{Q}^{\text{init}})$ over $\mathcal{F}$. Since $E[R + \gamma(\pi \hat{Q}^{\text{init}})(S') \mid S, A] = \mathcal{T}(\hat{Q}^{\text{init}})(S, A)$, replacing the conditional
937 Bellman target by the noisy target changes the risk only by an additive term that does not depend on
938 Q . Hence $\mathcal{T}_\mathcal{F}(\hat{Q}^{\text{init}})$ also minimizes $\hat{R}_0$. $\square$

939 The following results control the empirical process fluctuations that arise in the analysis of the inexact
940 Picard iteration. Recall the critical radius

$$\delta_n := \inf \left\{ \delta > 0 : \frac{\mathcal{J}(\delta, \mathcal{F})}{\sqrt{n} \delta^2} \leq 1 \right\}.$$

941 **Lemma 9.** *Let w be a state-action function with $\|w\|_\infty \leq B$, where $B \geq 1$. Assume that R , $\hat{Q}^{\text{init}}$,
942 $\mathcal{T}_\mathcal{F}(\hat{Q}^{\text{init}})$, and $\hat{Q}_n$ are uniformly bounded by M . Assume further that Conditions C5 and C6 hold.
943 Then there exists a constant $C = C(M) > 0$ such that, for all $\tau \in (0, 1)$, with probability at least
944 $1 - \tau$,*

$$\begin{aligned} & \left| (P_n - P_0) \left[w(\mathcal{T}_\mathcal{F}(\hat{Q}^{\text{init}}) - \hat{Q}_n) \{R + \gamma(\pi \hat{Q}^{\text{init}}) - \hat{Q}_n\} \right] \right| \\ & \leq C \left(B \delta_n^2 + \delta_n \|w(\mathcal{T}_\mathcal{F}(\hat{Q}^{\text{init}}) - \hat{Q}_n)\| \right) \\ & \quad + C \left(\frac{\sqrt{\log(1/\tau)} \|w(\mathcal{T}_\mathcal{F}(\hat{Q}^{\text{init}}) - \hat{Q}_n)\|}{\sqrt{n}} + \frac{B \log(1/\tau)}{n} \right), \end{aligned}$$

945 where $\|\cdot\|$ denotes the behavior-distribution norm $L^2(P_0)$.

946 *Proof.* Fix a bounded weight function w with $\|w\|_\infty \leq B$. Conditional on the sample used to
947 estimate w , this function is fixed. Let

$$\mathcal{V}_\mathcal{F} := \{ \pi Q : Q \in \mathcal{F} \}, \quad (\pi Q)(s) := \sum_{a \in \mathcal{A}} \pi(a \mid s) Q(s, a).$$

948 Define the function class

$$\begin{aligned} \mathcal{G}_w := & \left\{ (s, a, r, s') \mapsto w(s, a) (Q_1(s, a) - Q_2(s, a)) \right. \\ & \left. \times (r + \gamma V_1(s') - Q_2(s, a)) : Q_1, Q_2 \in \mathcal{F}, V_1 \in \mathcal{V}_\mathcal{F} \right\}. \end{aligned}$$

949 By boundedness, each $g \in \mathcal{G}_w$ satisfies

$$|g(s, a, r, s')| \leq C(M) |w(s, a) \{Q_1(s, a) - Q_2(s, a)\}|,$$

950 and hence

$$\|g\|_{L^2(P_0)} \leq C(M) \|w(Q_1 - Q_2)\|_{L^2(P_0)}. \quad (5)$$

951 **Entropy control.** It remains only to check that $\mathcal{V}_{\mathcal{F}}$ does not require a new entropy condition. For
 952 any discrete distribution Q_S on S , let $\bar{Q}$ be the lifted state–action distribution

$$\bar{Q}(s, a) := Q_S(s)/|\mathcal{A}|.$$

953 For any $Q, \tilde{Q} \in \mathcal{F}$, the target policy is fixed, so

$$|(\pi Q)(s) - (\pi \tilde{Q})(s)| = \left| \sum_{a \in \mathcal{A}} \pi(a | s) \{Q(s, a) - \tilde{Q}(s, a)\} \right| \leq |\mathcal{A}|^{1/2} \|Q(s, \cdot) - \tilde{Q}(s, \cdot)\|_{\ell_2(\text{Unif}(\mathcal{A}))}.$$

954 Therefore

$$\|\pi Q - \pi \tilde{Q}\|_{L^2(Q_S)} \leq |\mathcal{A}|^{1/2} \|Q - \tilde{Q}\|_{L^2(Q)}.$$

955 It follows that, for every $\varepsilon > 0$,

$$N(\varepsilon, \mathcal{V}_{\mathcal{F}}, L^2(Q_S)) \leq N(\varepsilon/|\mathcal{A}|^{1/2}, \mathcal{F}, L^2(\bar{Q})).$$

956 Since the entropy supremum for $\mathcal{F}$ ranges over all discrete state–action distributions, the uniform
 957 entropy of $\mathcal{V}_{\mathcal{F}}$ is controlled by that of $\mathcal{F}$, up to constants depending only on the finite action space.

958 The map $(Q_1, Q_2, V_1) \mapsto w(Q_1 - Q_2)(r + \gamma V_1 - Q_2)$ is a Lipschitz-continuous pointwise transfor-
 959 mation of uniformly bounded functions with a fixed bounded multiplier w ; the multiplier envelope B
 960 is kept explicit in the maximal inequality below. Standard permanence properties of entropy under
 961 Lipschitz maps and bounded multipliers [79] imply that there is a constant $C_1 = C_1(M, |\mathcal{A}|)$ such
 962 that, for all $\delta > 0$,

$$\mathcal{J}(\delta, \mathcal{G}_w) \leq C_1 \mathcal{J}(C_1 \delta, \mathcal{F}).$$

963 Thus $\mathcal{G}_w$ has the same critical radius as $\mathcal{F}$, up to constants absorbed into C .

964 Let

$$\mathcal{H}_w := \{tg : g \in \mathcal{G}_w, t \in [-1, 1]\}$$

965 be the symmetric star-shaped hull. The same entropy bound holds for $\mathcal{H}_w$. By Condition C6,

$$\frac{\mathcal{J}(\delta, \mathcal{F})}{\delta \sqrt{\log \log(1/\delta)}} \rightarrow \infty \quad \text{as } \delta \downarrow 0,$$

966 so $n^{-1/2} \sqrt{\log \log(1/\delta_n)} = o(\delta_n)$. Lemmas 7 and 6 applied to $\mathcal{H}_w$ give that, with probability at
 967 least $1 - \tau$, every $g \in \mathcal{G}_w$ satisfies

$$\begin{aligned} |(P_n - P_0)g| &\leq C \left(B\delta_n^2 + \delta_n \|g\|_{L^2(P_0)} \right) \\ &\quad + C \left(\frac{\sqrt{\log(1/\tau)} \|g\|_{L^2(P_0)}}{\sqrt{n}} + \frac{B \log(1/\tau)}{n} \right). \end{aligned}$$

968 Taking $Q_1 = \mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})$, $Q_2 = \hat{Q}_n$, and $V_1 = \pi \hat{Q}^{\text{init}}$, and using (5), yields the displayed bound. $\square$

969 **Main bound for inexact Picard iteration.** The following lemma bounds the regret incurred by an
 970 inexact Picard iteration in terms of the critical radius of $\mathcal{F}$ and the bounded-weight coverage constant.
 971 It allows the starting point $\hat{Q}^{\text{init}}$ to be an arbitrary (possibly data-dependent) element of $\mathcal{F}$.

972 Define the weighted norm:

$$\|f\|_{2, w_\pi}^2 := P_0[w_\pi f^2].$$

973 **Lemma 10** (Admissible-weight one-step regression error). Assume $\hat{Q}^{\text{init}} \in \mathcal{F}$, $|R| \leq M$,
 974 $\|\hat{Q}^{\text{init}}\|_\infty \leq M$, $\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})\|_\infty \leq M$, and $\|\hat{Q}_n\|_\infty \leq M$. Assume further that Conditions C8,
 975 C3, C5, C6, and C7 hold. Then there exists a constant $C = C(M) > 0$ such that, for all $\tau \in (0, 1)$,
 976 with probability at least $1 - \tau$,

$$\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_{2, w_\pi} \leq \frac{C}{1 - \rho_{\mathcal{H}}} \left\{ \kappa_{\text{cov}} \left(\delta_n + \sqrt{\frac{\log(1/\tau)}{n}} \right) + \omega_{\text{Bell}, w} \right\}.$$

977 *Proof of Lemma 10.* The proof first uses the empirical first-order condition, then lower bounds
 978 the corresponding population term by the estimated-weight curvature minus a Bellman-residual
 979 interaction. We then control the empirical-process fluctuation and solve the resulting scalar inequality.

980 By first-order optimality of $\hat{Q}_n = \arg \min_{Q \in \mathcal{F}} \hat{R}_n(Q)$ and convexity of $\mathcal{F}$, we have, for all $Q \in \mathcal{F}$,

$$\begin{aligned} & \frac{1}{n} \sum_{i=1}^n \hat{w}_\pi(S_i, A_i) \{Q(S_i, A_i) - \hat{Q}_n(S_i, A_i)\} \\ & \quad \times \{R_i + \gamma(\pi \hat{Q}^{\text{init}})(S'_i) - \hat{Q}_n(S_i, A_i)\} \leq 0. \end{aligned}$$

981 Since $\mathcal{T}_\mathcal{F}(\hat{Q}^{\text{init}}) \in \mathcal{F}$, taking $Q = \mathcal{T}_\mathcal{F}(\hat{Q}^{\text{init}})$ gives

$$\begin{aligned} & \frac{1}{n} \sum_{i=1}^n \hat{w}_\pi(S_i, A_i) \{\mathcal{T}_\mathcal{F}(\hat{Q}^{\text{init}})(S_i, A_i) - \hat{Q}_n(S_i, A_i)\} \\ & \quad \times \{R_i + \gamma(\pi \hat{Q}^{\text{init}})(S'_i) - \hat{Q}_n(S_i, A_i)\} \leq 0. \end{aligned}$$

982 Adding and subtracting the population expectation and rearranging,

$$\begin{aligned} & P_0[\hat{w}_\pi(\mathcal{T}_\mathcal{F}(\hat{Q}^{\text{init}}) - \hat{Q}_n) \{R + \gamma(\pi \hat{Q}^{\text{init}}) - \hat{Q}_n\}] \\ & \leq (P_0 - P_n)[\hat{w}_\pi(\mathcal{T}_\mathcal{F}(\hat{Q}^{\text{init}}) - \hat{Q}_n) \{R + \gamma(\pi \hat{Q}^{\text{init}}) - \hat{Q}_n\}]. \end{aligned} \tag{6}$$

983 Here we abuse notation and write R for the coordinate projection $(s, a, r, s') \mapsto r$, and similarly view
 984 $(\pi \hat{Q}^{\text{init}})$ as the mapping $(s, a, r, s') \mapsto (\pi \hat{Q}^{\text{init}})(s')$.

985 Let

$$d := \mathcal{T}_\mathcal{F}(\hat{Q}^{\text{init}}) - \hat{Q}_n, \quad b := \mathcal{T}(\hat{Q}^{\text{init}}) - \mathcal{T}_\mathcal{F}(\hat{Q}^{\text{init}}).$$

986 Applying the law of total expectation gives

$$P_0[\hat{w}_\pi d \{R + \gamma(\pi \hat{Q}^{\text{init}}) - \hat{Q}_n\}] = P_0[\hat{w}_\pi d(d + b)].$$

987 Because $d \in \mathcal{H}_\mathcal{F}$, Condition C7 implies

$$P_0[\hat{w}_\pi d^2] \geq (1 - \rho_\mathcal{H}) \|d\|_{2, w_\pi}^2.$$

988 Moreover, the projection-equivalence condition implies $P_0[w_\pi db] \geq 0$. Therefore, by Cauchy–
 989 Schwarz and the definition of $\omega_{\text{Bell}, w}$,

$$P_0[\hat{w}_\pi db] = P_0[w_\pi db] + P_0[(\hat{w}_\pi - w_\pi)db] \geq -\omega_{\text{Bell}, w} \|d\|_{2, w_\pi}.$$

990 Combining these inequalities yields

$$P_0[\hat{w}_\pi d \{R + \gamma(\pi \hat{Q}^{\text{init}}) - \hat{Q}_n\}] \geq (1 - \rho_\mathcal{H}) \|d\|_{2, w_\pi}^2 - \omega_{\text{Bell}, w} \|d\|_{2, w_\pi}.$$

991 Applying this lower bound to (6) and rearranging,

$$\begin{aligned} & (1 - \rho_\mathcal{H}) \|\mathcal{T}_\mathcal{F}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_{2, w_\pi}^2 \\ & \leq (P_0 - P_n)[\hat{w}_\pi(\mathcal{T}_\mathcal{F}(\hat{Q}^{\text{init}}) - \hat{Q}_n) \{R + \gamma(\pi \hat{Q}^{\text{init}}) - \hat{Q}_n\}] \\ & \quad + \omega_{\text{Bell}, w} \|\mathcal{T}_\mathcal{F}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_{2, w_\pi}. \end{aligned} \tag{7}$$

992 Denote the regression residual by

$$\xi_n := R + \gamma(\pi \hat{Q}^{\text{init}}) - \hat{Q}_n.$$

993 The empirical-process term decomposes as

$$\begin{aligned} & (P_0 - P_n)[\hat{w}_\pi(\mathcal{T}_\mathcal{F}(\hat{Q}^{\text{init}}) - \hat{Q}_n) \xi_n] \\ & = (P_0 - P_n)[w_\pi(\mathcal{T}_\mathcal{F}(\hat{Q}^{\text{init}}) - \hat{Q}_n) \xi_n] \\ & \quad + (P_0 - P_n)[(\hat{w}_\pi - w_\pi)(\mathcal{T}_\mathcal{F}(\hat{Q}^{\text{init}}) - \hat{Q}_n) \xi_n]. \end{aligned}$$

994 We now invoke Condition C5 and apply Lemma 9 conditionally on the data used to train the weights
 995 to the two terms on the right-hand side, with $w = w_\pi$ and $w = \hat{w}_\pi - w_\pi$. Condition C3 ensures that
 996 these multipliers have envelope bounded by a constant multiple of κ_{cov} . For

$$\Delta := \mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n \in \mathcal{H}_{\mathcal{F}},$$

997 Condition C3 also gives the norm comparisons

$$\|w_\pi \Delta\|_{L^2(P_0)} \leq \kappa_{\text{cov}} \|\Delta\|_{2, w_\pi}, \quad \|(\hat{w}_\pi - w_\pi) \Delta\|_{L^2(P_0)} \leq \kappa_{\text{cov}} \|\Delta\|_{2, w_\pi}.$$

998 Hence, for $C = C(M) > 0$, for all $\tau \in (0, 1)$, with probability at least $1 - \tau$,

$$\begin{aligned} & \left| (P_0 - P_n) [\hat{w}_\pi (\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n) \{R + \gamma(\pi \hat{Q}^{\text{init}}) - \hat{Q}_n\}] \right| \\ & \leq C \kappa_{\text{cov}} \left(\delta_n^2 + \delta_n \|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_{2, w_\pi} \right) \\ & \quad + C \kappa_{\text{cov}} \left(\frac{\sqrt{\log(1/\tau)} \|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_{2, w_\pi}}{\sqrt{n}} + \frac{\log(1/\tau)}{n} \right). \end{aligned}$$

999 Combining this high-probability bound with (7), and writing

$$x := \|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_{2, w_\pi}, \quad a := \delta_n + \sqrt{\log(1/\tau)/n}, \quad \omega := \omega_{\text{Bell}, w},$$

1000 gives

$$(1 - \rho_{\mathcal{H}})x^2 \leq C \kappa_{\text{cov}}(a^2 + ax) + \omega x.$$

1001 Solving this scalar inequality and using $\kappa_{\text{cov}} \geq 1$ and $\rho_{\mathcal{H}} < 1$ yields

$$x \leq \frac{C}{1 - \rho_{\mathcal{H}}} \{\kappa_{\text{cov}} a + \omega\},$$

1002 which is the claim. □

1003 G Proofs of Main Results

1004 *Proof of Theorem 1.* Write

$$\begin{aligned} Q_{\mathcal{F}}^\pi - Q^\pi &= \mathcal{T}_{\mathcal{F}} Q_{\mathcal{F}}^\pi - \mathcal{T} Q^\pi \\ &= (\mathcal{T}_{\mathcal{F}} Q_{\mathcal{F}}^\pi - \mathcal{T}_{\mathcal{F}} Q^\pi) + (\mathcal{T}_{\mathcal{F}} Q^\pi - \mathcal{T} Q^\pi). \end{aligned}$$

1005 Taking $L^2(\mu_\pi)$ norms and applying the triangle inequality

$$\|Q_{\mathcal{F}}^\pi - Q^\pi\|_{2, \mu_\pi} \leq \|\mathcal{T}_{\mathcal{F}} Q_{\mathcal{F}}^\pi - \mathcal{T}_{\mathcal{F}} Q^\pi\|_{2, \mu_\pi} + \|\mathcal{T}_{\mathcal{F}} Q^\pi - \mathcal{T} Q^\pi\|_{2, \mu_\pi}.$$

1006 By the contraction property of Lemma 5,

$$\|\mathcal{T}_{\mathcal{F}} Q_{\mathcal{F}}^\pi - \mathcal{T}_{\mathcal{F}} Q^\pi\|_{2, \mu_\pi} \leq \gamma \|Q_{\mathcal{F}}^\pi - Q^\pi\|_{2, \mu_\pi},$$

1007 and by definition of $\mathcal{T}_{\mathcal{F}}$,

$$\mathcal{T}_{\mathcal{F}} Q^\pi - \mathcal{T} Q^\pi = \Pi_{\mathcal{F}} \mathcal{T} Q^\pi - \mathcal{T} Q^\pi = -(I - \Pi_{\mathcal{F}}) \mathcal{T} Q^\pi.$$

1008 Combining these two displays yields

$$\|Q_{\mathcal{F}}^\pi - Q^\pi\|_{2, \mu_\pi} \leq \gamma \|Q_{\mathcal{F}}^\pi - Q^\pi\|_{2, \mu_\pi} + \|(I - \Pi_{\mathcal{F}}) \mathcal{T} Q^\pi\|_{2, \mu_\pi}.$$

1009 Rearranging, we obtain

$$(1 - \gamma) \|Q_{\mathcal{F}}^\pi - Q^\pi\|_{2, \mu_\pi} \leq \|(I - \Pi_{\mathcal{F}}) \mathcal{T} Q^\pi\|_{2, \mu_\pi},$$

1010 which proves the first inequality:

$$\|Q_{\mathcal{F}}^\pi - Q^\pi\|_{2, \mu_\pi} \leq \frac{1}{1 - \gamma} \|(I - \Pi_{\mathcal{F}}) \mathcal{T} Q^\pi\|_{2, \mu_\pi}.$$

1011 For the second inequality, note that $\Pi_{\mathcal{F}} \mathcal{T} Q^\pi$ is the best $L^2(\mu_\pi)$ approximation to $\mathcal{T} Q^\pi$ in $\mathcal{F}$, so

$$\|(I - \Pi_{\mathcal{F}}) \mathcal{T} Q^\pi\|_{2, \mu_\pi} = \inf_{f \in \mathcal{F}} \|\mathcal{T} Q^\pi - f\|_{2, \mu_\pi}.$$

1012 Since Q^π is a fixed point of $\mathcal{T}$, we have $\mathcal{T} Q^\pi = Q^\pi$, and hence

$$\|(I - \Pi_{\mathcal{F}}) \mathcal{T} Q^\pi\|_{2, \mu_\pi} = \inf_{f \in \mathcal{F}} \|Q^\pi - f\|_{2, \mu_\pi}.$$

1013 Substituting this into the previous display yields

$$\|Q_{\mathcal{F}}^\pi - Q^\pi\|_{2, \mu_\pi} \leq \frac{1}{1 - \gamma} \inf_{f \in \mathcal{F}} \|f - Q^\pi\|_{2, \mu_\pi},$$

1014 which completes the proof. $\square$

1015 *Proof of Lemma 2.* By Lemma 5, $\mathcal{T}_{\mathcal{F}}$ is a γ -contraction on $L^2(\mu_\pi)$. Let

$$a_k := \|\hat{Q}^{(k)} - Q_{\mathcal{F}}^\pi\|_{2, \mu_\pi}, \quad k \geq 0.$$

1016 For any $k \geq 1$,

$$\begin{aligned} a_k &= \|\hat{Q}^{(k)} - Q_{\mathcal{F}}^\pi\|_{2, \mu_\pi} \\ &\leq \|\hat{Q}^{(k)} - \mathcal{T}_{\mathcal{F}}(\hat{Q}^{(k-1)})\|_{2, \mu_\pi} + \|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{(k-1)}) - Q_{\mathcal{F}}^\pi\|_{2, \mu_\pi} \\ &\leq \eta_k + \|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{(k-1)}) - \mathcal{T}_{\mathcal{F}}(Q_{\mathcal{F}}^\pi)\|_{2, \mu_\pi} \\ &\leq \eta_k + \gamma \|\hat{Q}^{(k-1)} - Q_{\mathcal{F}}^\pi\|_{2, \mu_\pi} = \eta_k + \gamma a_{k-1}, \end{aligned}$$

1017 where we used the assumed approximation bound in the first term, the fixed-point property $\mathcal{T}_{\mathcal{F}} Q_{\mathcal{F}}^\pi =$
1018 $Q_{\mathcal{F}}^\pi$ in the second line, and the contraction property in the third.

1019 Thus the sequence $\{a_k\}$ satisfies the recursion

$$a_k \leq \gamma a_{k-1} + \eta_k, \quad k \geq 1.$$

1020 Unrolling this recursion yields

$$\begin{aligned} a_K &\leq \gamma^K a_0 + \sum_{j=1}^K \gamma^{K-j} \eta_j \\ &= \gamma^K \|\hat{Q}^{(0)} - Q_{\mathcal{F}}^\pi\|_{2, \mu_\pi} + \sum_{j=1}^K \gamma^{K-j} \eta_j, \end{aligned}$$

1021 which gives the desired bound. $\square$

1022 *Proof of Lemma 3.* In the main-text specialization, $w_\pi = d\mu_\pi/d\nu_b$. Hence $w_\pi \nu_b = \mu_\pi$, so
1023 Condition C8 holds because the weighted and stationary projection objectives are identical, and
1024 the $L^2(w_\pi \nu_b)$ and $L^2(\mu_\pi)$ norms coincide. Therefore Lemma 10 applies to the k th regression
1025 with $\hat{Q}^{\text{init}} = \hat{Q}^{(k)}$ and $\hat{Q}_n = \hat{Q}^{(k+1)}$, giving the stated stationary-norm bound after using
1026 $\omega_{\text{Bell}, w}(\hat{Q}^{(k)}) \leq \omega_{\text{Bell}, w}$. $\square$

1027 *Proof of Theorem 2.* We prove the result by applying the one-step regression bound uniformly over
1028 the K fitted regressions and then unrolling the deterministic inexact Picard recursion. By Lemma 3
1029 and a union bound, there exists a constant

$$C = C(M) > 0$$

1030 such that, with probability at least $1 - \tau$, for all $k \in [K]$,

$$\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{(k-1)}) - \hat{Q}^{(k)}\|_{2, \mu_\pi} \leq \frac{C}{1 - \rho_{\mathcal{H}}} \left\{ \kappa_{\text{cov}} \left(\delta_n + \sqrt{\frac{\log(K/\tau)}{n}} \right) + \omega_{\text{Bell}, w} \right\}.$$

1031 Define

$$\varepsilon := \frac{C}{1 - \rho_{\mathcal{H}}} \left\{ \kappa_{\text{cov}} \left(\delta_n + \sqrt{\frac{\log(K/\tau)}{n}} \right) + \omega_{\text{Bell}, w} \right\},$$

1032 so that on this event,

$$\|\hat{Q}^{(k)} - \mathcal{T}_{\mathcal{F}}(\hat{Q}^{(k-1)})\|_{2, \mu_{\pi}} \leq \varepsilon \quad \text{for all } k \in [K].$$

1033 Thus, by Lemma 2, for any $K \geq 1$,

$$\|\hat{Q}^{(K)} - Q_{\mathcal{F}}^{\pi}\|_{2, \mu_{\pi}} \leq \gamma^K \|\hat{Q}^{(0)} - Q_{\mathcal{F}}^{\pi}\|_{2, \mu_{\pi}} + \sum_{j=1}^K \gamma^{K-j} \varepsilon.$$

1034 The geometric sum is bounded by $(1 - \gamma)^{-1}$, which gives the displayed bound. $\square$

1035 **Corollary 1** (Finite-sample bound under admissible weights). *Assume Conditions C1–C7 and, in*
 1036 *addition, Conditions C8 and C9. Then there exists $C = C(M, c) < \infty$ such that, for all $\tau \in (0, 1)$*
 1037 *and $K \in \mathbb{N}$, with probability at least $1 - \tau$,*

$$\|\hat{Q}^{(K)} - Q_{\mathcal{F}}^{\pi}\|_{2, \mu_{\pi}} \leq \gamma^K \|\hat{Q}^{(0)} - Q_{\mathcal{F}}^{\pi}\|_{2, \mu_{\pi}} + \frac{C}{(1 - \gamma)(1 - \rho_{\mathcal{H}})} \{\varepsilon_{\text{stat}}(K, \tau) + \omega_{\text{Bell}, w}\}.$$

1038 *Proof.* Apply Lemma 10 to each of the K weighted regressions and take a union bound. Condition C9
 1039 converts the resulting $L^2(w_{\pi} \nu_b)$ one-step error to $L^2(\mu_{\pi})$, with the factor c absorbed into C . Lemma 2
 1040 then unrolls the same inexact Picard recursion used in the proof of Theorem 2. $\square$

1041 *Proof of Theorem 3.* We bound the distance between the fixed points of the projected Bellman
 1042 operators induced by r_0 and by $\hat{r}$. By Lemma 5 and stationarity of μ_{π} , $\mathcal{T}_{\mathcal{F}}$ is a γ -contraction on
 1043 $L^2(\mu_{\pi})$.

1044 By definition of the fixed points,

$$Q_{\mathcal{F}}^{\pi} = \mathcal{T}_{\mathcal{F}}(Q_{\mathcal{F}}^{\pi}), \quad \hat{Q}_{\mathcal{F}}^{\pi} = \hat{\mathcal{T}}_{\mathcal{F}}(\hat{Q}_{\mathcal{F}}^{\pi}).$$

1045 Write

$$\hat{Q}_{\mathcal{F}}^{\pi} - Q_{\mathcal{F}}^{\pi} = \hat{\mathcal{T}}_{\mathcal{F}}(\hat{Q}_{\mathcal{F}}^{\pi}) - \mathcal{T}_{\mathcal{F}}(Q_{\mathcal{F}}^{\pi}),$$

1046 and add and subtract $\mathcal{T}_{\mathcal{F}}(\hat{Q}_{\mathcal{F}}^{\pi})$:

$$\hat{Q}_{\mathcal{F}}^{\pi} - Q_{\mathcal{F}}^{\pi} = \underbrace{(\hat{\mathcal{T}}_{\mathcal{F}}(\hat{Q}_{\mathcal{F}}^{\pi}) - \mathcal{T}_{\mathcal{F}}(\hat{Q}_{\mathcal{F}}^{\pi}))}_{(I)} + \underbrace{(\mathcal{T}_{\mathcal{F}}(\hat{Q}_{\mathcal{F}}^{\pi}) - \mathcal{T}_{\mathcal{F}}(Q_{\mathcal{F}}^{\pi}))}_{(II)}.$$

1047 Taking $L^2(\mu_{\pi})$ -norms and using the triangle inequality,

$$\begin{aligned} \|\hat{Q}_{\mathcal{F}}^{\pi} - Q_{\mathcal{F}}^{\pi}\|_{2, \mu_{\pi}} &\leq \|\hat{\mathcal{T}}_{\mathcal{F}}(\hat{Q}_{\mathcal{F}}^{\pi}) - \mathcal{T}_{\mathcal{F}}(\hat{Q}_{\mathcal{F}}^{\pi})\|_{2, \mu_{\pi}} \\ &\quad + \|\mathcal{T}_{\mathcal{F}}(\hat{Q}_{\mathcal{F}}^{\pi}) - \mathcal{T}_{\mathcal{F}}(Q_{\mathcal{F}}^{\pi})\|_{2, \mu_{\pi}}. \end{aligned}$$

1048 The second term is controlled by the contraction property:

$$\|\mathcal{T}_{\mathcal{F}}(\hat{Q}_{\mathcal{F}}^{\pi}) - \mathcal{T}_{\mathcal{F}}(Q_{\mathcal{F}}^{\pi})\|_{2, \mu_{\pi}} \leq \gamma \|\hat{Q}_{\mathcal{F}}^{\pi} - Q_{\mathcal{F}}^{\pi}\|_{2, \mu_{\pi}}.$$

1049 Hence

$$(1 - \gamma) \|\hat{Q}_{\mathcal{F}}^{\pi} - Q_{\mathcal{F}}^{\pi}\|_{2, \mu_{\pi}} \leq \|\hat{\mathcal{T}}_{\mathcal{F}}(\hat{Q}_{\mathcal{F}}^{\pi}) - \mathcal{T}_{\mathcal{F}}(\hat{Q}_{\mathcal{F}}^{\pi})\|_{2, \mu_{\pi}}. \quad (8)$$

1050 Next, observe that for any Q ,

$$\begin{aligned} \|\hat{\mathcal{T}}_{\mathcal{F}}(Q) - \mathcal{T}_{\mathcal{F}}(Q)\|_{2, \mu_{\pi}} &= \|\Pi_{\mathcal{F}} \hat{\mathcal{T}}(Q) - \Pi_{\mathcal{F}} \mathcal{T}(Q)\|_{2, \mu_{\pi}} \\ &\leq \|\hat{\mathcal{T}}(Q) - \mathcal{T}(Q)\|_{2, \mu_{\pi}} \\ &= \|\hat{r} - r_0\|_{2, \mu_{\pi}}, \end{aligned}$$

1051 since $\hat{\mathcal{T}}(Q) - \mathcal{T}(Q) = \hat{r} - r_0$ does not depend on Q . Substituting into (8) yields

$$\|\hat{Q}_{\mathcal{F}}^{\pi} - Q_{\mathcal{F}}^{\pi}\|_{2, \mu_{\pi}} \leq \frac{1}{1 - \gamma} \|\hat{r} - r_0\|_{2, \mu_{\pi}}.$$

1052 **Stronger guarantee under linear function classes.** If $\mathcal{F}$ is a linear subspace of $L^2(\mu_\pi)$, then $\Pi_{\mathcal{F}}$
 1053 is a linear orthogonal projection, and we obtain a sharper reward misspecification bound. In this case,
 1054 for any Q ,

$$\begin{aligned}\|\hat{\mathcal{T}}_{\mathcal{F}}(Q) - \mathcal{T}_{\mathcal{F}}(Q)\|_{2, \mu_\pi} &= \|\Pi_{\mathcal{F}}\hat{\mathcal{T}}(Q) - \Pi_{\mathcal{F}}\mathcal{T}(Q)\|_{2, \mu_\pi} \\ &= \|\Pi_{\mathcal{F}}(\hat{\mathcal{T}}(Q) - \mathcal{T}(Q))\|_{2, \mu_\pi} \\ &= \|\Pi_{\mathcal{F}}(\hat{r} - r_0)\|_{2, \mu_\pi},\end{aligned}$$

1055 since $\hat{\mathcal{T}}(Q) - \mathcal{T}(Q) = \hat{r} - r_0$ does not depend on Q . In particular,

$$\hat{\mathcal{T}}_{\mathcal{F}}(\hat{Q}_{\mathcal{F}}^\pi) - \mathcal{T}_{\mathcal{F}}(\hat{Q}_{\mathcal{F}}^\pi) = \Pi_{\mathcal{F}}(\hat{r} - r_0),$$

1056 so

$$\|\hat{\mathcal{T}}_{\mathcal{F}}(\hat{Q}_{\mathcal{F}}^\pi) - \mathcal{T}_{\mathcal{F}}(\hat{Q}_{\mathcal{F}}^\pi)\|_{2, \mu_\pi} = \|\Pi_{\mathcal{F}}(\hat{r} - r_0)\|_{2, \mu_\pi}.$$

1057 Substituting this into (8) yields

$$\|\hat{Q}_{\mathcal{F}}^\pi - Q_{\mathcal{F}}^\pi\|_{2, \mu_\pi} \leq \frac{1}{1 - \gamma} \|\Pi_{\mathcal{F}}(\hat{r} - r_0)\|_{2, \mu_\pi},$$

1058 so only the component of the reward error lying in $\mathcal{F}$ affects the fixed point.

1059 □

1060 H Convergence of the Stationary-Ratio Saddle Estimator

1061 This section gives a finite-dimensional convergence result for a minimax stationary state-action
 1062 density-ratio estimator. A useful quantity for bounding the residual interaction in Theorem 2 is the
 1063 relative weight error

$$\varepsilon_{\text{wt}} := \|\hat{w}_\pi / w_\pi - 1\|_{2, w_\pi \nu_b}.$$

1064 The result allows both ratio-class and critic-class approximation error.

1065 **Relation to existing minimax theory.** The estimator is closest to minimax weight learning for
 1066 off-policy evaluation [70]. Those bounds control value error through Bellman-moment residuals,
 1067 complexity terms, and approximation error, but do not directly imply the $L^2(\nu_b)$ ratio error needed
 1068 here without an additional identification condition. This distinction between projected moment error
 1069 and true L^2 error is central in minimax conditional-moment estimation [8]. We therefore follow
 1070 a two-step argument: first prove projected moment convergence, then impose approximate critic
 1071 identifiability to obtain L^2 ratio convergence. Related fast-rate and RKHS extensions appear in
 1072 Olivas-Martinez and Rotnitzky [59], Uehara et al. [71].

1073 **Setup.** Let $X = (S, A) \in \mathcal{X} := \mathcal{S} \times \mathcal{A}$. The ratio-estimation sample consists of i.i.d. draws
 1074 $Z_i = (X_i, X_i^+)$, where $X_i \sim \nu_b$, $S'_i \sim P(\cdot \mid S_i, A_i)$, $A'_i \sim \pi(\cdot \mid S'_i)$, and $X_i^+ = (S'_i, A'_i)$.
 1075 Assume the Markov chain induced by (π, P) has a unique stationary state-action distribution μ_π ,
 1076 with $\mu_\pi \ll \nu_b$, and define $w_\pi = d\mu_\pi / d\nu_b$. For any $g : \mathcal{X} \rightarrow \mathbb{R}$, write $\Delta g(Z) := g(X) - g(X^+)$.
 1077 The stationary ratio satisfies

$$E_{\nu_b}\{w_\pi(X)\Delta g(Z)\} = 0 \quad \text{for all bounded measurable } g, \quad E_{\nu_b}\{w_\pi(X)\} = 1. \quad (9)$$

1078 **Proposition 1** (Population identification). *Let $w \geq 0$ be measurable with $E_{\nu_b}\{w(X)\} = 1$. If*
 1079 *$E_{\nu_b}[w(X)\{g(X) - g(X^+)\}] = 0$ for all bounded measurable g , then $w \cdot \nu_b$ is stationary for (π, P) .*
 1080 *If the stationary distribution is unique, then $w = w_\pi$, ν_b -almost surely.*

1081 *Proof.* Let λ be the probability measure with density w relative to ν_b . The moment condition implies

$$\int g(x) d\lambda(x) = \int E\{g(X^+) \mid X = x\} d\lambda(x) \quad \text{for all bounded measurable } g.$$

1082 Thus one transition from initial law λ leaves all bounded test-function expectations unchanged, so λ
 1083 is invariant. By uniqueness, $\lambda = \mu_\pi$, hence $w = w_\pi$ ν_b -almost surely. □

1084 **Estimator.** Let $\psi : \mathcal{X} \rightarrow \mathbb{R}^p$ be ratio features and $\phi : \mathcal{X} \rightarrow \mathbb{R}^m$ be critic features. Define $w_\theta(x) :=$
1085 $\theta^\top \psi(x)$, $g_\beta(x) := \beta^\top \phi(x)$, and $\Delta\phi(Z) := \phi(X) - \phi(X^+)$. For $\Theta \subseteq \mathbb{R}^p$, let $\mathcal{W} = \{w_\theta : \theta \in \Theta\}$
1086 and $\mathcal{G} = \{g_\beta : \beta \in \mathbb{R}^m\}$. With $P_n f = n^{-1} \sum_{i=1}^n f(Z_i)$, define

$$\widehat{\theta} \in \arg \min_{\theta \in \Theta} \sup_{\beta \in \mathbb{R}^m} \left\{ P_n[w_\theta(X)\beta^\top \Delta\phi(Z)] - \frac{1}{2} P_n[(\beta^\top \phi(X))^2] \right\} + \frac{\kappa}{2} \{P_n w_\theta - 1\}^2 + \frac{\lambda}{2} \|\theta\|_2^2, \quad (10)$$

1087 and set $\widehat{w}_\pi = w_{\widehat{\theta}}$. Let

$$\widehat{M} := P_n[\Delta\phi(Z)\psi(X)^\top], \quad \widehat{\Sigma} := P_n[\phi(X)\phi(X)^\top], \quad \widehat{b} := P_n[\psi(X)],$$

1088 and let M, Σ, b denote their population counterparts. If $\widehat{\Sigma}$ is positive definite, the inner supremum
1089 equals $\frac{1}{2} \|\widehat{M}\theta\|_{\widehat{\Sigma}^{-1}}^2$, so $\widehat{\theta}$ minimizes

$$\widehat{J}_\lambda(\theta) = \frac{1}{2} \|\widehat{M}\theta\|_{\widehat{\Sigma}^{-1}}^2 + \frac{\kappa}{2} (\widehat{b}^\top \theta - 1)^2 + \frac{\lambda}{2} \|\theta\|_2^2.$$

1090 The population projected moment discrepancy is

$$\mathcal{M}(\theta)^2 := \|M\theta\|_{\Sigma^{-1}}^2 + \kappa (b^\top \theta - 1)^2.$$

1091 **Assumptions.** We use the following finite-dimensional conditions.

1092 *Assumption 1* (Sampling). The sample $\{Z_i\}_{i=1}^n$ is i.i.d. The ratio-estimation sample is independent
1093 of the Bellman-regression sample, or is made conditionally independent by sample splitting or
1094 cross-fitting.

1095 *Assumption 2* (Bounded linear classes). The set Θ is closed and convex. There exist
1096 $B_\psi, B_\phi, B_d, B_\Theta < \infty$ such that, almost surely,

$$\|\psi(X)\|_2 \leq B_\psi, \quad \|\phi(X)\|_2 \leq B_\phi, \quad \|\Delta\phi(Z)\|_2 \leq 2B_\phi, \quad \Theta \subseteq \{\theta : \|\theta\|_2 \leq B_\Theta\},$$

1097 and $0 \leq w_\theta(X) \leq B_d$ for all $\theta \in \Theta$.

1098 *Assumption 3* (Critic Gram conditioning). Σ is positive definite and $\lambda_{\min}(\Sigma) \geq \sigma_\phi > 0$.

1099 *Assumption 4* (Overlap). There exist $0 < \underline{d} \leq \bar{d} < \infty$ such that

$$\underline{d} \leq w_\pi(X) \leq \bar{d} \quad \nu_b\text{-almost surely.}$$

1100 **Definition 1** (Oracle approximation errors). For any $\theta^\dagger \in \Theta$, define

$$\eta_D(\theta^\dagger) := \|w_{\theta^\dagger} - w_\pi\|_{2, \nu_b}, \quad \eta_M(\theta^\dagger) := \|M\theta^\dagger\|_{\Sigma^{-1}} + \sqrt{\kappa} |b^\top \theta^\dagger - 1|.$$

1101 Since w_π satisfies (9),

$$M\theta^\dagger = E_{\nu_b}[(w_{\theta^\dagger}(X) - w_\pi(X))\Delta\phi(Z)], \quad b^\top \theta^\dagger - 1 = E_{\nu_b}[w_{\theta^\dagger}(X) - w_\pi(X)].$$

1102 Thus bounded features imply $\eta_M(\theta^\dagger) \leq C_M \eta_D(\theta^\dagger)$, where C_M depends only on $B_\phi, \sigma_\phi, \kappa$.

1103 *Assumption 5* (Approximate critic identifiability). Fix $\theta^\dagger \in \Theta$ and let $\Gamma := E_{\nu_b}[\psi(X)\psi(X)^\top]$. There
1104 exist $\zeta > 0$ and $\eta_G \geq 0$ such that, for every $\theta \in \Theta$,

$$\|w_\theta - w_{\theta^\dagger}\|_{2, \nu_b} \leq \frac{1}{\zeta} \{ \|M(\theta - \theta^\dagger)\|_{\Sigma^{-1}} + \sqrt{\kappa} |b^\top (\theta - \theta^\dagger)| \} + \eta_G.$$

1105 *Remark 3* (Interpretation). The saddle objective directly controls only critic-projected stationarity
1106 moments and normalization. Assumption 5 says that these moments identify the ratio over Θ , up to
1107 critic approximation slack η_G . The exact restricted-eigenvalue case is $\eta_G = 0$, where

$$h^\top (M^\top \Sigma^{-1} M + \kappa b b^\top) h \geq \zeta^2 h^\top \Gamma h \quad \text{for all } h \in \Theta - \theta^\dagger.$$

1108 Without such a compatibility condition, the saddle objective yields projected moment convergence
1109 but not the L^2 ratio error required by Theorem 2.

1110 **Projected moment convergence.** Let $r_n(\delta) := \sqrt{\{p + m + \log(1/\delta)\}/n}$.

1111 **Lemma 11** (Finite-dimensional concentration). *Under Assumptions 1–3, there exist $c, C < \infty$,
1112 depending only on the boundedness and conditioning constants, such that, with probability at least
1113 $1 - \delta$,*

$$\|\widehat{\Sigma} - \Sigma\|_{\text{op}} + \|\widehat{M} - M\|_{\text{op}} + \|\widehat{b} - b\|_2 \leq Cr_n(\delta).$$

1114 *If $r_n(\delta) \leq c$, then $\widehat{\Sigma}$ is positive definite and $\|\widehat{\Sigma}^{-1} - \Sigma^{-1}\|_{\text{op}} \leq Cr_n(\delta)$.*

1115 *Proof.* Each summand in $\widehat{b} - b$, $\widehat{M} - M$, and $\widehat{\Sigma} - \Sigma$ is bounded. Vector and matrix Bernstein
1116 inequalities, together with a union bound, give the concentration bound. The inverse perturbation
1117 bound follows from Weyl's inequality and $\lambda_{\min}(\Sigma) \geq \sigma_\phi$. $\square$

1118 **Theorem 5** (Projected moment convergence). *Suppose Assumptions 1–3 hold. Fix $\theta^\dagger \in \Theta$. There
1119 exist $c, C < \infty$, depending only on the boundedness and conditioning constants, such that if
1120 $r_n(\delta) \leq c$, then with probability at least $1 - \delta$, every solution $\widehat{\theta}$ of (10) satisfies*

$$\mathcal{M}(\widehat{\theta}) \leq C \left\{ \eta_M(\theta^\dagger) + r_n(\delta) + \sqrt{\lambda} \|\theta^\dagger\|_2 \right\}.$$

1121 *Proof.* Let

$$\widehat{\mathcal{M}}(\theta)^2 := \|\widehat{M}\theta\|_{\widehat{\Sigma}^{-1}}^2 + \kappa(\widehat{b}^\top \theta - 1)^2.$$

1122 On the event in Lemma 11, boundedness of Θ gives

$$\sup_{\theta \in \Theta} |\widehat{\mathcal{M}}(\theta) - \mathcal{M}(\theta)| \leq C_1 r_n(\delta).$$

1123 By optimality of $\widehat{\theta}$,

$$\frac{1}{2} \widehat{\mathcal{M}}(\widehat{\theta})^2 + \frac{\lambda}{2} \|\widehat{\theta}\|_2^2 \leq \frac{1}{2} \widehat{\mathcal{M}}(\theta^\dagger)^2 + \frac{\lambda}{2} \|\theta^\dagger\|_2^2.$$

1124 Dropping the nonnegative term on the left and taking square roots gives

$$\widehat{\mathcal{M}}(\widehat{\theta}) \leq \widehat{\mathcal{M}}(\theta^\dagger) + \sqrt{\lambda} \|\theta^\dagger\|_2.$$

1125 Applying the uniform comparison at $\widehat{\theta}$ and $\theta^\dagger$ yields the claim. $\square$

1126 **L^2 ratio convergence.** Projected moment convergence is the strongest conclusion available without
1127 an identification condition. The next theorem converts projected moment error into true $L^2(\nu_b)$ ratio
1128 error.

1129 **Theorem 6** (Finite-dimensional L^2 convergence). *Suppose Assumptions 1–5 hold for an oracle
1130 $\theta^\dagger \in \Theta$. There exist $c, C < \infty$, depending only on the boundedness, conditioning, and κ , such that if
1131 $r_n(\delta) \leq c$, then with probability at least $1 - \delta$, every solution $\widehat{\theta}$ of (10) satisfies*

$$\|\widehat{w}_\pi - w_\pi\|_{2, \nu_b} \leq C \left[\eta_D(\theta^\dagger) + \frac{1}{\zeta} \left\{ \eta_M(\theta^\dagger) + r_n(\delta) + \sqrt{\lambda} \|\theta^\dagger\|_2 \right\} + \eta_G \right].$$

1132 Since $\eta_M(\theta^\dagger) \leq C_M \eta_D(\theta^\dagger)$,

$$\|\widehat{w}_\pi - w_\pi\|_{2, \nu_b} \leq C' \left[\left(1 + \frac{1}{\zeta} \right) \eta_D(\theta^\dagger) + \eta_G + \frac{1}{\zeta} \left\{ r_n(\delta) + \sqrt{\lambda} \|\theta^\dagger\|_2 \right\} \right].$$

1133 If $\lambda \lesssim r_n(\delta)^2$, the statistical term has rate $r_n(\delta) = \sqrt{\{p + m + \log(1/\delta)\}/n}$.

1134 *Proof.* By Assumption 5,

$$\|\widehat{w}_\pi - w_{\theta^\dagger}\|_{2, \nu_b} \leq \frac{1}{\zeta} \left\{ \|M(\widehat{\theta} - \theta^\dagger)\|_{\Sigma^{-1}} + \sqrt{\kappa} |b^\top (\widehat{\theta} - \theta^\dagger)| \right\} + \eta_G.$$

1135 The expression in braces is at most $\mathcal{M}(\widehat{\theta}) + \eta_M(\theta^\dagger)$. Applying Theorem 5 and then the triangle
1136 inequality

$$\|\widehat{w}_\pi - w_\pi\|_{2, \nu_b} \leq \|\widehat{w}_\pi - w_{\theta^\dagger}\|_{2, \nu_b} + \eta_D(\theta^\dagger)$$

1137 proves the first display. The second follows from $\eta_M(\theta^\dagger) \leq C_M \eta_D(\theta^\dagger)$. $\square$

1138 **Translation to the FQE relative-error norm.** The residual-interaction sufficient bounds use the
 1139 relative error

$$\varepsilon_{\text{wt}} = \|\widehat{w}_\pi/w_\pi - 1\|_{2, w_\pi \nu_b}.$$

1140 The following corollary converts the previous $L^2(\nu_b)$ bound into this norm.

1141 **Corollary 2** (Weight-error bound for stationary-weighted FQE). *Suppose the conditions of Theorem 6*
 1142 *and Assumption 4 hold. Then*

$$\varepsilon_{\text{wt}} \leq \underline{d}^{-1/2} \|\widehat{w}_\pi - w_\pi\|_{2, \nu_b}.$$

1143 *Consequently, with probability at least $1 - \delta$,*

$$\varepsilon_{\text{wt}} \leq C \underline{d}^{-1/2} \left[\eta_D(\theta^\dagger) + \frac{1}{\zeta} \left\{ \eta_M(\theta^\dagger) + \sqrt{\frac{p+m+\log(1/\delta)}{n}} + \sqrt{\lambda} \|\theta^\dagger\|_2 \right\} + \eta_G \right].$$

1144 *If $\eta_M(\theta^\dagger) \leq C_M \eta_D(\theta^\dagger)$ and $\lambda \lesssim \{p+m+\log(1/\delta)\}/n$, then*

$$\varepsilon_{\text{wt}} \leq C' \underline{d}^{-1/2} \left[\left(1 + \frac{1}{\zeta}\right) \eta_D(\theta^\dagger) + \eta_G + \frac{1}{\zeta} \sqrt{\frac{p+m+\log(1/\delta)}{n}} \right].$$

1145 *Thus the right-hand side can be substituted for ε_w in the residual-interaction sufficient bounds*
 1146 *following Theorem 2.*

1147 *Proof.* By definition of the $w_\pi \nu_b$ -weighted norm,

$$\|\widehat{w}_\pi/w_\pi - 1\|_{2, w_\pi \nu_b}^2 = \int \frac{(\widehat{w}_\pi(x) - w_\pi(x))^2}{w_\pi(x)} d\nu_b(x) \leq \underline{d}^{-1} \|\widehat{w}_\pi - w_\pi\|_{2, \nu_b}^2.$$

1148 Taking square roots and applying Theorem 6 proves the result. $\square$

1149 **Scope.** The saddle objective directly controls $\mathcal{M}(\widehat{\theta})$, a projected stationarity-moment discrepancy.
 1150 True $L^2(\nu_b)$ ratio convergence requires approximate critic identifiability. The term $\eta_D(\theta^\dagger)$ is ratio-
 1151 class approximation error, while η_G measures failure of the critic class to identify ratio perturbations
 1152 through stationary-flow moments.